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Design and Implementation of a Low-Power LoRa Wireless Sensor Mesh Network with Cloud Storage for Monitoring Soil Conditions on Icelandic Turf Roofs

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60 ECTS thesis submitted in partial fulfillment of a
Magister Scientiarum degree in Electrical and Computer Engineering

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Reykjavik, August 2026

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(Low-Power LoRa Mesh Network for Icelandic Turf Roofs)

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Abstract

The thesis presents the design, implementation, and evaluation of a low-power wireless sensor mesh network intended for monitoring soil and environmental conditions beneath Icelandic turf roofs. The system uses low-cost, off-the-shelf sensor nodes based on an Arduino Pro Mini, an SX1278-based 433 MHz LoRa transceiver, a capacitive soil moisture sensor, a DS18B20 soil temperature sensor, an AHT10 temperature and humidity sensor, local SD-card storage, a real-time clock, and a three-cell 18650 battery pack. A LoRa base station forwards collected data to the ThingSpeak cloud platform for remote access and visualization.

Laboratory and field testing evaluated power consumption, wireless communication, sensor performance, soil moisture calibration, and cloud upload reliability. Under the documented worst-case operating cycle, the sensor node consumed approximately 3.08 mA on average, corresponding to an idealized lifetime of approximately 142 days with a 10,500 mAh battery capacity. Range testing around Vellir in Hafnarfjörður achieved reliable single-hop communication over an approximate 3 km radius using correctly matched 433 MHz LoRa antennas. Long-term deployment at Keldur demonstrated continuous environmental monitoring, although two nodes experienced hardware failures near the end of the documented deployment period. Soil moisture calibration produced a nonlinear second-order relationship over the 0–40% moisture range, with an approximate root mean square error of 3% soil moisture relative to the HOBO reference. ThingSpeak uploads using HTTP POST achieved an average upload time of approximately 1–2 seconds and an observed failure rate of 0%.

The results demonstrate the feasibility of a low-cost, low-power LoRa-based monitoring system for Icelandic turf roofs and provide a practical foundation for longer-term heritage-site environmental monitoring.

Keywords: wireless sensor network (WSN), LoRa, mesh network, soil moisture, turf roofs, environmental monitoring, low-power systems, ThingSpeak

Útdráttur

TODO: Add the Icelandic abstract before final submission.

Acknowledgments

TODO: Finalize the acknowledgments.

About the Author

TODO: Finalize the author biography.

Abbreviations

AH	Absolute Humidity
FDR	Frequency Domain Reflectometry
RH	Relative Humidity
TDR	Time Domain Reflectometry
WSN	Wireless Sensor Network
IoT	Internet of Things
LoRa	Long Range
RTC	Real-Time Clock
MCU	Microcontroller Unit
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
ADC	Analog-to-Digital Converter
VWC	Volumetric Water Content
RSSI	Received Signal Strength Indicator
PCB	Printed Circuit Board
SD	Secure Digital
I^2C	Inter-Integrated Circuit
SPI	Serial Peripheral Interface
API	Application Programming Interface
BW	Bandwidth
SF	Spreading Factor
CR	Coding Rate
CRC	Cyclic Redundancy Check
PL	Payload Length
OOP	Object-Oriented Programming
IDE	Integrated Development Environment
HTTP	Hypertext Transfer Protocol
GPIO	General-Purpose Input/Output
NTP	Network Time Protocol
RX	Receive
TX	Transmit
ISM	Industrial, Scientific, and Medical
RMSE	Root Mean Square Error
USB	Universal Serial Bus
LED	Light-Emitting Diode
Li-Ion	Lithium-Ion

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1. Introduction

1.1. Definition of Terms

1.1.1. Turf House

A turf house is defined as a structure or building whose roof and walls are mainly made from turf. Sometimes, these can be mixed with gravel or stones for extra support. Timber beams and rafters are used as the main structure of the building and are also used to support and hold the roof. Turf is also used as a main component to cover the roof, and slates are used as an underlay. Throughout the years, the material used for turf houses has not changed drastically, but the form of these houses has changed and adapted through the ages [1][2].

1.1.2. Soil Moisture

Soil moisture is defined as the amount or quantity of water in the soil [3][4]. It can be further defined as the amount of water held in between the pores of the soil. These volumes of water may be in liquid or vaporized form [5]. When measuring the amount of water in a known amount of soil, soil moisture can be expressed as "% water by weight or volume of soil", or "inches of water per foot of soil" [6].

1.1.3. Relative Humidity in Soil

Humidity is defined as the amount of gaseous water or water vapor in the air. High humidity means there is a lot of moisture in the air and low humidity indicates dry conditions. Humidity is often described as absolute humidity (AH) or relative humidity (RH) [7][8]. Absolute humidity is measured in units of g/m^3 where it is the measure of water vapor content in air that is not dependent on temperature [9]. Relative humidity is the ratio of the current amount of water vapor in the air to the

maximum amount the air can hold at a given temperature, expressed as a percentage (%) [10].

This study will collect data and conduct tests below the soil surface. In this context, humidity refers to the relative humidity of the air within the soil's pores and gaps, which will be used as data for the study.

1.2. Background of the Study

Icelandic turf houses are important cultural and architectural heritage structures whose turf construction requires continued maintenance. The literature supplied for this thesis describes the long history of turf construction in Iceland and the need to understand turf layers and their environmental conditions. Moisture is particularly relevant because turf and timber components are susceptible to deterioration, while the conservation of heritage structures requires maintenance approaches that do not unnecessarily disturb the site.

The current system design addresses this monitoring problem by placing low-power sensor nodes beneath turf roofs. The methodology specifies that nodes collect soil moisture, soil temperature, air temperature, and relative humidity, while transmitting measurements through a LoRa mesh network to a base station and cloud database. The nodes are designed to remain visually unobtrusive and to operate for extended periods with limited maintenance.

1.3. Statement of the Problem

Monitoring environmental conditions beneath turf roofs presents several engineering constraints. Sensor nodes must be sufficiently small and unobtrusive to avoid affecting the appearance of a heritage site, while also operating in an environment characterized by high moisture, freezing and thawing cycles, and possible snow coverage. The system must provide reliable wireless communication, conserve battery energy, retain measurements when transmissions are missed, and make the collected data available remotely.

The project therefore addresses the need for a practical monitoring system that combines low-cost sensing hardware, low-power operation, long-range wireless communication, local data backup, and cloud-based access in a single deployable platform.

1.4. Objectives

The study aims to design, implement, and evaluate a low-power LoRa wireless sensor mesh network for monitoring soil and environmental conditions beneath Icelandic turf roofs. The specific objectives are to:

1. measure soil moisture, soil temperature, air temperature, relative humidity, and battery voltage using distributed sensor nodes;

2. develop a LoRa-based mesh communication system capable of reliably transferring measurements between sensor nodes and a base station;

3. implement low-power operating modes suitable for long-duration battery-powered deployment;

4. provide local data backup so measurements are retained when wireless transmissions are missed;

5. calibrate the low-cost capacitive soil moisture sensor against reference measurements; and

6. transmit collected measurements to a cloud platform for remote access, visualization, and analysis.

1.5. Significance of the Study

The project provides a low-cost engineering approach to continuous monitoring of environmental conditions on Icelandic turf roofs. The system is intended to support researchers and conservation personnel by providing measurements without requiring frequent manual sampling or intrusive installation. The use of off-the-shelf components and a modular mesh architecture also provides a basis for scaling the monitoring approach to additional turf-roof sites.

1.6. Scope and Limitations

The work focuses on a low-power LoRa-based wireless sensor mesh network using distributed sensor nodes and a cloud-connected base station. The implemented nodes measure soil moisture, soil temperature, air temperature, relative humidity, and

battery voltage. The methodology specifies burial of sensor nodes at approximately 11 cm, weatherproof enclosures, local SD-card data backup, and a non-destructive installation approach.

The evaluation includes laboratory power-consumption testing, preliminary and intermediate field deployments around the University of Iceland, LoRa range testing around Vellir in Hafnarfjörður, long-term deployment at the Keldur turf-house site, soil moisture calibration, and ThingSpeak cloud-upload testing. The documented results also show limitations, including antenna-related packet loss during early testing, sensor-calibration issues during intermediate testing, and hardware failures affecting two nodes near the end of the long-term deployment. The calibration results cover the 0–40% moisture range and therefore do not establish performance beyond that range.

2. Review of Related Literature

2.1. Icelandic Turf Houses

Turf houses are a significant part of culture and heritage in Iceland. A paper published by Bathurst et al. [11], explains that turf has always been widely used across regions in northern Europe for its durability, renewability, and its abundance. It also has great insulation properties which makes it a good choice for colder climates. Since turf houses are unique, their appearance and construction garnered a lot of attention and admiration from people. This leads to interesting studies and research on their history, culture and construction practices [12].

According to Lin et al. [13], regions such as Scotland and especially Iceland have very limited supply of natural resources, however turf was abundant. This made it very important for shelter construction. Turf houses that were well built were also stated to be better in insulating cold weather than houses built on stone and timber, which is very important in these countries with harsh cold weather. Wunderlich et al. [14], describes turf as a major component in buildings for Iceland during the medieval period. The turf used in the construction of these buildings was cut from local bogs or from edges of cultivated fields in the area. For example, in Mosfellsbær, it was presumed that the bricks of turf used were cut from the valley floor bogs, which was south of the habitation area. The bogs were then transported to higher elevation areas, where the difference in elevation naturally aided in draining water, making it suitable for constructing houses [11].

Stone was often laid below the turf walls to prevent moisture from seeping into the walls and eventually causing it to rot [15]. The amount of stone beneath the turf walls varied on use. For example, in turf houses that sheltered animals, the stone layers were higher so that it would prevent animals from damaging the turf by feeding on it or rubbing against it. Archaeological studies suggest that in Iceland specifically, the stone foundations were used less in buildings during the Viking era when compared to later eras such as the Medieval and Post Medieval [14].

Like any other natural resource with positive qualities, turf also has its drawbacks. A book published by Nannini [12], explains that insulation with turf was only tolerable

since it was applied generously, houses would have thick walls and almost completely no windows. Additionally, settlers would burn dried manure inside these houses to generate more heat. Combined with the lack of ventilation, this led to very unhealthy environments. Another issue is that turf is replenishable. This means settlers would have to renovate every few years or so and force owners to refurbish or rebuild their home. Hannes Larusson [16] emphasizes this as "an inseparable part of the turf house" which he later states as one of the main reasons why they were eventually replaced by concrete.

With the innovation of concrete and the eventual abandonment of refurbishing or maintaining these perishable turf houses, this led to the loss and destruction of these historical buildings.

Studies in Turf and Turf House Maintenance

Due to the unique architectural design of these turf houses, it became one of the most researched topics on Icelandic architectural history. Studies even dating back to the last decades of the nineteenth century [12]. Even today, many scholars and authors research on turf construction from analysis of inner environment comfort to botany of the turf layers [17]. According to Sigurjón [18], many Icelandic scholars and researchers such as architects, architectural historians and conservation experts, have been extensively studying turf houses solely focusing on restoring and conserving these sites.

Researchers such as Wunderlich et al. [14], conducted studies on imaging medieval turf buildings in Iceland through geophysical methods. This is to understand the structural features and subsurface characteristics of the houses. During a study conducted by Milek and Roberts [19], ash was discovered to be deposited on the floors of the houses. This was used as insecticide and a drying material since the ash can fill the tiny gaps and holes. It is also absorbent and hence can protect the wooden materials from fungal decay. Studies from Milek [20] state that although turf buildings can last for over 100 years when properly constructed and maintained, these buildings would still need constant turf replacement. Specifically on the roof and walls, since they are prone to rot. Stating that both turf and timber needs replacement every 10 – 20 years.

There has also been other research with goals specifically focused on turf restoration in general. Studies conducted by Hnatowich et al. [21] have been transplanting healthy turf into disturbed areas with hopes to restore it. By using native and local turf for the transplant, there is an added benefit of lesser shock to the transplanted area. Additionally, other studies such as those by Malleshaiah et al. [22], have presented techniques such as soil aerification. This has proved to improve turf quality

by relieving the soil compaction, giving it proper aeration and therefore improving turf quality. This is further supported by Nunome et al. [23] stating that soil compacting is a major factor in the increase of degradation of turf.

With ongoing and continuous studies and discoveries on turf and turf house construction and maintenance. Preservation experts can further study and utilize these techniques to extend the lifespan of these turf houses. By minimizing rot and decay on these turf and wooden structures, especially on the roof and walls. These turf houses can be properly maintained and can last for lifetimes. Preserving the cultural heritage and importance of these historical structures.

2.2. Measuring Soil Moisture

Measuring soil moisture is one of the fundamental techniques when studying soil conditions in environmental monitoring research. Over the years, soil monitoring techniques have been continuously improved, and new methods have been developed. However, these methods vary in their principles and specific applications.

The standard method of measuring soil moisture is the Gravimetric Method. To measure the soil moisture, researchers must get the difference of the natural weight of soil to the same amount of soil but dried. This is often achieved by placing the soil in an oven to remove moisture [24]. Starr and Paltineanu [25] state this method as very simple and highly accurate to such a degree that it is often used as a calibration for other sensors or methods. It is even used to validate data gathered from different studies [26][27]. The formula for calculating the soil moisture is defined by:

$$Soil\ Moisture\ Content\ (\%) = \frac{Weight\ of\ Wet\ Soil - Weight\ of\ Dry\ Soil}{Weight\ of\ Dry\ Soil}\ x\ 100 \tag{2.1}$$

However, this method has its downsides, although being very simple. It is very time-consuming and destructive for the study area since it involves unearthing soil and taking the time to dehydrate it in the oven. This makes it unsuitable for long term studies and real time monitoring [24].

New and modern techniques that are highly accurate, fast and non-destructive are the Time Domain Reflectometry (TDR) and the Frequency Domain Reflectometry (FDR). Because there is a significant difference between the dielectric constants of dry soil and pure water, modern tools can easily distinguish between the two [28]. TDR works by placing steel rods into the soil and then sending electrical signals through

them. The signal travels from one probe through the other by traveling through the soil. The property of wet soil makes it return the signal slower compared to dry soil. In turn, this helps the device determine the humidity of the soil accurately [6][29].

FDR works similarly to TDR, but instead of measuring the time it takes for the signal to move through the soil, FDR measures the frequency of the electromagnetic waves when it travels [28]. When the electromagnetic waves travel through the soil, the frequency of the signal changes based on the moisture. This helps researchers determine the soil moisture accurately and precisely [30]. Although these methods are quick and accurate without having the need to be destructive, devices like these are expensive and can cost up to hundreds or even thousands of dollars based on the application [6].

Nowadays, there are newer types of sensors which can be used to measure soil moisture without having the need to be expensive all while being fairly accurate, quick, and non-destructive. These sensors include resistance-based and capacitance-based types. Resistance-based sensors measure soil moisture by placing probes into the soil, similar to the methods mentioned earlier. However, instead of sending electrical signals or waves, this method measures the soil's resistance. The resistance of the soil lowers when it has a high amount of moisture and rises when it is dry [31][32]. However, Koyunco et al. [28] states that this method is highly unreliable because soil types can vary in resistance which leads to imprecise results. Additionally, the probes are prone to corrosion which can lead to no results at all.

Capacitive sensors work by measuring the capacitance across the sensor electrode due to humidity when deploying a pulse voltage [33][34]. González-Turel et al. [35] explain that because the dielectric constant of water is higher than that of air or dry soil, the moisture content affects the dielectric constant. This results in an increase when the soil is wet, leading to a higher capacitance that can be detected by the sensor. Researchers like Kulmány et al. [36] prefer these types of sensors because they provide accurate data while being inexpensive and easy to install. Additionally, these sensors are less prone to oxidation and damage because they do not require exposed probes to touch the soil; their sealed design provides adequate protection.

Among most methods and sensors used for determining soil moisture, typically capacitive sensors are often preferred for their accuracy, inexpensive price, and ease of installation while being able to provide real time data. Each method, however, has its own strength and limitations, which are considered during research making one more suitable than the others, especially for specific conditions. Ultimately, the choice of sensor and methods will depend on the specific requirements and parameters in the study and the choice of the researchers. It's important to evaluate these factors and select the one that is most appropriate for the study.

3. Methodology

3.1. Overview

This chapter describes the system design and presents the methodology used to design and develop the project while achieving the objectives of the study. It outlines the complete workflow of the project, starting from the system concept and design requirements, through hardware and firmware development, PCB design and manufacturing, and finally to system deployment. The chapter also describes the calibration and validation of the selected soil moisture sensor in comparison with professional-class sensors.

3.2. System Concept and Design Requirements

3.2.1. System Concept

The system is designed as a wireless mesh network using LoRa as the core communication technology. Weatherproof wireless sensor nodes collect data such as soil moisture, soil temperature, air humidity, and air temperature, and transmit this data efficiently between neighboring nodes until it reaches a base station with access to a cloud database. The collected data is then made accessible to researchers from anywhere.

LoRa was selected for its effectiveness over the short to medium distances between turf houses, with the mesh topology improving network robustness.

3.2.2. Functional Requirements

Since turf houses are considered natural heritage sites and museums, the system must be invisible to visitors. Therefore, the sensor nodes must be well hidden while

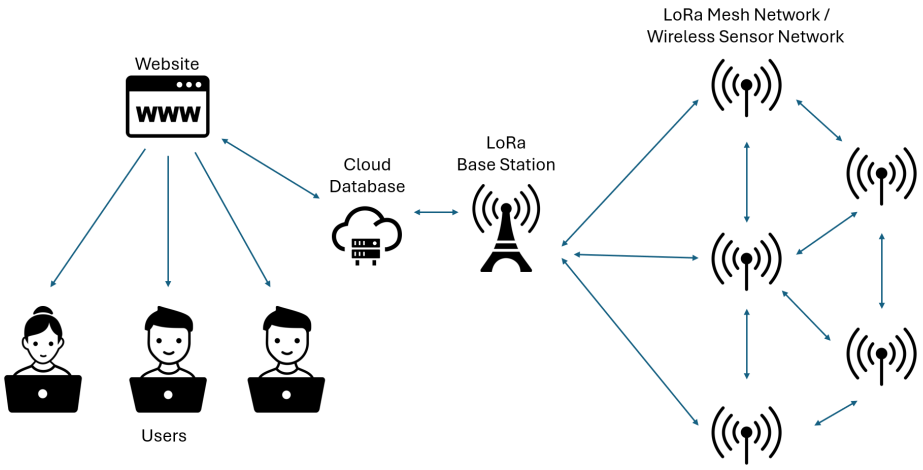


Figure 3.1: System concept of the LoRa-based wireless sensor mesh network for monitoring turf roof conditions.

still being effective in gathering data. The functional requirements of the system are as follows:

- The sensor nodes must be capable of accurately measuring the required data, including soil moisture, soil temperature, air temperature, and air humidity.
- The system must reliably transmit data through the mesh network, even with disturbances between nodes and the presence of noise.
- The system must be able to operate for extended periods of time, as maintenance will be sparse due to the remote locations of turf houses.
- The sensor nodes must be able to back up data to local storage in case of missed transmissions.
- The collected data must be uploaded to a cloud network where it can be accessed and analyzed by researchers from anywhere.

3.2.3. Environmental and Design Constraints

Multiple design constraints were identified following discussions with experts in soil moisture sensing and turf roof construction. These discussions highlighted several

environmental and design factors that needed to be considered for the system. The identified constraints include the following:

- Sensor nodes will be buried at depths of approximately 11 cm and may be covered by additional layers of snow during winter.
- The device must be sufficiently weatherproof to withstand harsh Icelandic conditions, including high moisture levels from rain and exposure to repeated freezing and thawing cycles across seasons, as well as prolonged snow coverage.
- The installation and overall design process must be non-destructive to preserve the turf roof and must remain visually unobtrusive in order to maintain the integrity of the heritage site.

3.3. Hardware Design

3.3.1. In Situ Sensor

The sensor nodes deployed beneath the turf roofs consist of a low-power microcontroller unit (MCU), lithium-ion batteries, a real-time clock (RTC) for synchronizing data collection with the main network, an SD card for local data backup in case of missed transmissions, a soil moisture sensor, a temperature and humidity sensor, and a LoRa module for wireless communication.

Based on the scope of this project, low cost was one of the significant factors considered during system design. As a result, the system is primarily composed of off-the-shelf components that were carefully selected based on their cost and performance. Power consumption was another critical design consideration. To improve energy efficiency, all components were selected to operate at 3.3 V or lower.

To further reduce power consumption, the design incorporates a MOSFET-based power control system that disconnects power to sensors while the system is in sleep mode. The sensor node is periodically awakened by the real-time clock to collect, store, and transmit data before returning to low-power sleep operation.

Device Components

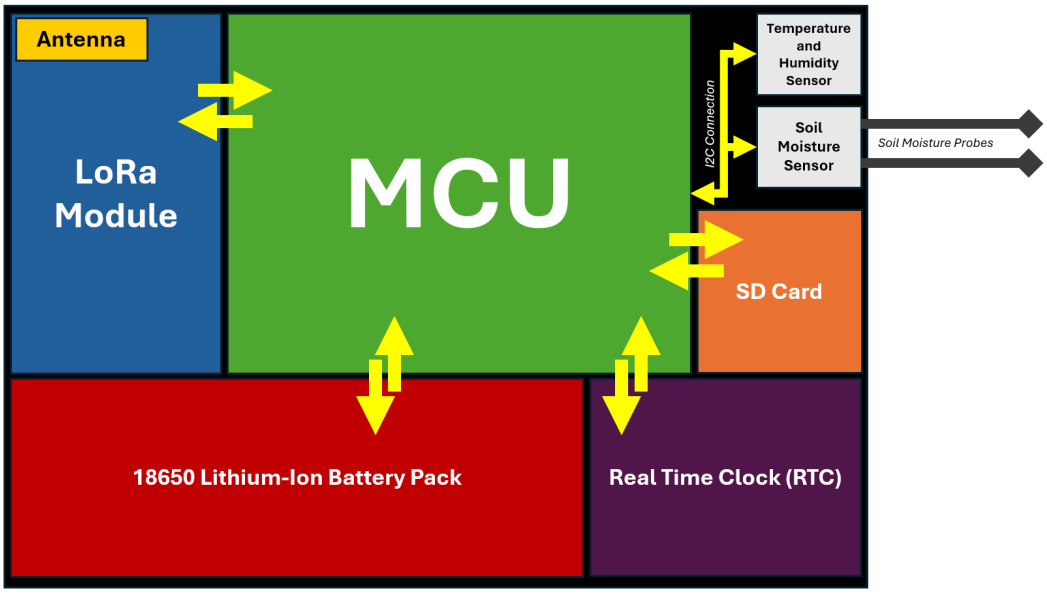


Figure 3.2: In-situ sensor node hardware configuration and component layout.

LoRa Module (Ra-02 SX1278 433Mhz)

The Ra-02 LoRa module, based on the SX1278 transceiver, operates at 433 MHz, which is within the legal ISM band in Iceland. The module offers long-range communication capabilities of up to several kilometers, which is more than sufficient for the expected distances between turf houses. Additionally, the module has low power consumption and a small physical footprint, making it suitable for a compact and energy-efficient sensor node design.

Microcontroller (Arduino Pro Mini)

The choice of microcontroller was a critical design decision. The microcontroller needed to be compact, energy efficient, and capable of handling sensor data acquisition, storage, and wireless communication. Several alternatives were considered, including the ESP32; however, it was deemed unsuitable due to higher power consumption, higher cost, and excess computational capability for the requirements of this project.

The Arduino Pro Mini, based on the ATmega328P microcontroller, was selected as it operates at 3.3 V with an 8 MHz clock frequency and provides 32 kB of flash memory. It supports both I2C and serial communication, which are required for interfacing with the sensors and peripheral modules. The available memory and processing capability were found to be sufficient for the application. Power consumption is approximately 4.5 mA during active operation and can be reduced

to approximately 3 µA in deep sleep mode after removing onboard indicator LEDs, which was implemented in this project.

Air Temperature and Humidity Sensor (AHT10)

The AHT10 sensor was selected for measuring air temperature and humidity due to its availability, low cost, and adequate accuracy for the application. The sensor communicates via the I2C interface and is capable of measuring temperatures from -40 °C to 85 °C with an accuracy of ± 0.3 °C. Relative humidity measurements range from 0% to 100% with an accuracy of $\pm 2\%$, which is sufficient for monitoring environmental conditions beneath turf roofs.

Soil Temperature Sensor (DS18B20)

The DS18B20 digital temperature sensor was selected for soil temperature measurements due to its simplicity, accuracy, and robustness. The sensor operates over a temperature range of -55 °C to 125 °C with an accuracy of ± 0.5 °C. It communicates using a one-wire interface, requiring only three connections, which simplifies wiring and reduces complexity. Waterproof probe-style versions of the DS18B20 are widely available, making the sensor well suited for buried soil applications.

Soil Moisture Sensor (Capacitive Soil Moisture Sensor v1.2)

A generic off-the-shelf capacitive soil moisture sensor (v1.2) was used in this study. The sensor design is open source and widely available, resulting in many similar commercial versions. This sensor outputs an analog voltage that varies with the soil moisture content, which is read by the Arduino as an analog value within the range defined by the 3.3 V supply voltage. Despite being low cost, previous studies have shown that capacitive soil moisture sensors can provide consistent and reliable measurements when properly selected and calibrated, as discussed in Chapter 2.

Capacitive sensors were chosen over resistance-based alternatives due to their improved reliability and reduced susceptibility to corrosion in moist soil environments. While professional-grade soil moisture measurement methods such as TDR and FDR offer higher accuracy, they are significantly more expensive and less suitable for large-scale deployment. In contrast, low-cost capacitive sensors offer a practical solution for distributed monitoring applications, such as turf roof installations.

In this study, the selected sensor was used with the understanding that calibration is required to improve measurement accuracy. By calibrating the sensor against professional-grade reference instruments, the system aims to achieve accurate soil moisture measurements while maintaining low cost and scalability.

SD Card Module (SparkFun OpenLog Serial Logger)

Local data storage was implemented using the SparkFun OpenLog serial logger module. Several SD card modules were initially evaluated; however, many were found to be too large, power intensive, or required software libraries that exceeded the available flash memory of the main microcontroller.

The OpenLog module records data received over a serial interface directly to a microSD card and is based on its own ATmega328 microcontroller, allowing it to operate independently of the main MCU. This approach significantly reduces memory usage on the sensor node. The module supports microSD cards up to 32 GB and consumes approximately 2–6 mA during operation, making it suitable for low-power data logging.

Power Source (18650 Samsung 35E 3500mAh Li-Ion Battery)

The system is powered using three 18650 lithium-ion batteries, each with a nominal capacity of 3500 mAh, providing a total capacity of 10,500 mAh. These batteries operate over a voltage range of approximately 2.65 V to 4.2 V, which is suitable for a 3.3 V system.

To ensure safe operation within voltage limits, a Zener diode is used to reduce the maximum input voltage by approximately 0.5 V. This configuration ensures compatibility with the 3.3 V power requirements while providing sufficient capacity for long-term operation.

Size was an important consideration in the design of the sensor modules. However, sourcing batteries with a small form factor and high capacity was severely limited by import fees, availability, and cost. As a result, a conventional and widely available 18650 lithium-ion battery was selected.

Real-Time Clock (DS3231)

The DS3231 real-time clock was selected due to its high accuracy, low power consumption, and integrated temperature-compensated crystal oscillator. The module includes a battery backup and consumes approximately 100 µA in standby mode. It supports up to two programmable alarms capable of generating an interrupt signal, which is used to wake the microcontroller from deep sleep mode.

This functionality is critical for synchronizing sensor nodes and ensuring consistent data collection and transmission intervals across the mesh network.

Antenna (433 MHz 3dbi)

The selected antenna operates at 433 MHz, which matches the configuration of the system. The antenna has a minimal design and is commonly used for radio communication applications, making it suitable for LoRa operation. Although the antenna has a gain of only 3 dBi, it provides significantly better performance than operating without an external antenna.

Additionally, the antenna's compact and discreet design allows it to blend well with the surrounding environment. Its small size and vertical form factor make it visually unobtrusive, resembling tall grass, which is important for maintaining the stealth and aesthetic requirements of the turf roof installation.

Schematic Design

This section describes the schematic design of the main printed circuit board (PCB) used for the sensor node. To compactly integrate all components into the smallest possible form factor and to simplify assembly and wiring, a custom PCB was designed and manufactured. The PCB houses all system components described in the previous section, with the exception of the batteries, along with additional circuitry required for system functionality.

Power Control and Battery Monitoring

Battery voltage monitoring is implemented using a simple voltage divider connected to analog pin A3 of the microcontroller. The microcontroller reads the divided voltage

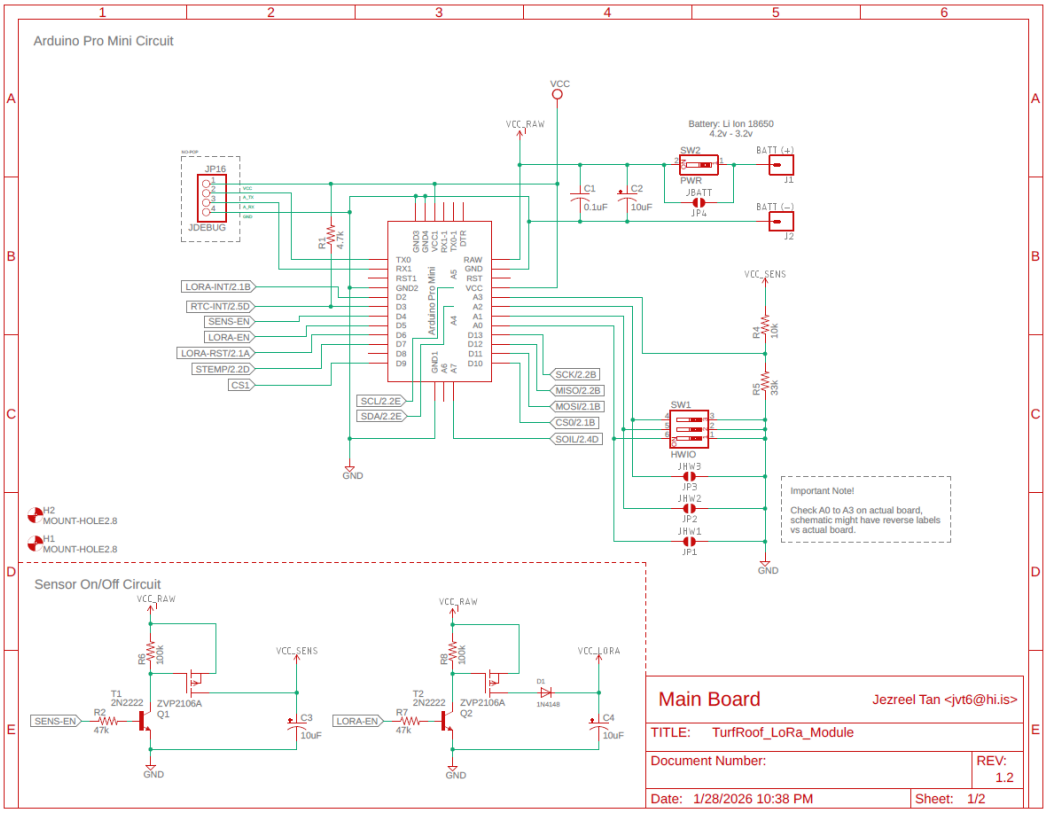


Figure 3.3: Battery voltage monitoring circuit and power-control design for the sensor node.

as an analog value, which is then converted back to the corresponding battery voltage in software. This allows the system to monitor battery health during operation.

To minimize power consumption during sleep periods, the design includes MOSFET-based power control for two sensor groups, labeled SENS-EN and LORA-EN. When the system enters deep sleep mode between data collection intervals, the microcontroller disables both MOSFETs, effectively cutting power to the sensors and the LoRa module. Each group is only powered when required and is turned off immediately afterward, ensuring minimal power consumption during inactive periods.

Hardware ID

As shown in the schematic, the core of the board is the Arduino Pro Mini microcontroller. A component labeled SW1 is used as a three-bit hardware identifier, allowing each sensor node to be assigned a unique ID. Upon boot, the microcontroller reads the state of these bits and assigns the corresponding hardware ID to the node, preventing data collisions during transmission. With this configuration, up to seven

unique devices can be supported within a single system. The design can be further scaled by adding additional identifier bits using available microcontroller pins. For deployments involving multiple systems, software-based hardware prefixes can be implemented to distinguish between different system groups.

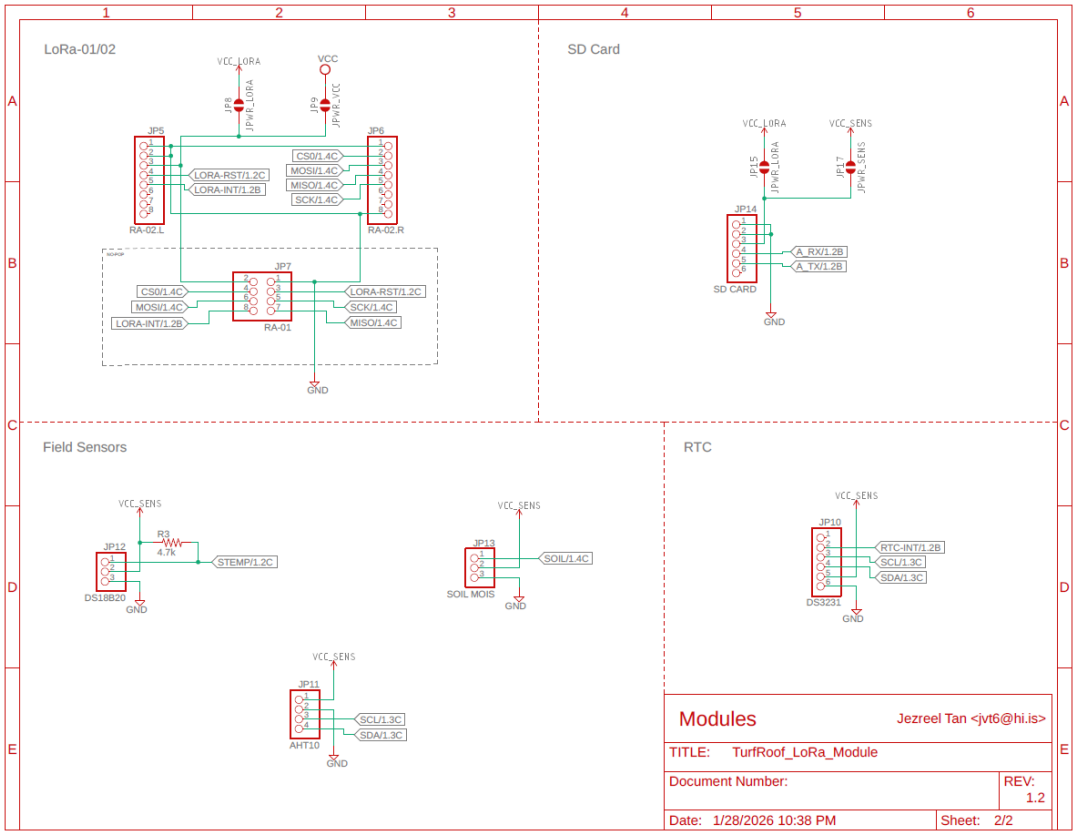


Figure 3.4: Custom PCB layout and component placement for the sensor node.

The PCB layout was primarily designed to accommodate the Ra-02 LoRa module. However, based on feedback from users with experience in LoRa systems, an alternative module package (Ra-01), which offers similar specifications but a different form factor, was also considered. To support this flexibility, a no-populate drop-in footprint was included on the PCB, allowing either module to be used depending on availability or user preference.

For the sensor interfaces, pin headers were used to simplify the design, testing, and development process. This approach allows sensors to be easily connected and disconnected, improving ease of assembly and enabling quicker replacement or modification during testing and deployment.

PCB Design and Manufacturing

During the PCB design process, cost and compactness were carefully considered. The main PCB for a single sensor module was designed to fit within a 50.5 mm × 50.5 mm footprint using a two-layer, 1.6 mm thick FR-4 substrate. A two-layer design was sufficient due to the relatively low routing complexity, helping to reduce manufacturing costs.

The PCB was manufactured by JLCPCB. Ground planes were incorporated where possible to improve signal integrity and reduce noise, particularly around the LoRa transceiver. Component placement was optimized to allow ease of manual assembly and debugging during development.

The use of pin headers for the sensors simplified the assembly process, allowing sensors to be rearranged easily and packed neatly within the enclosure.

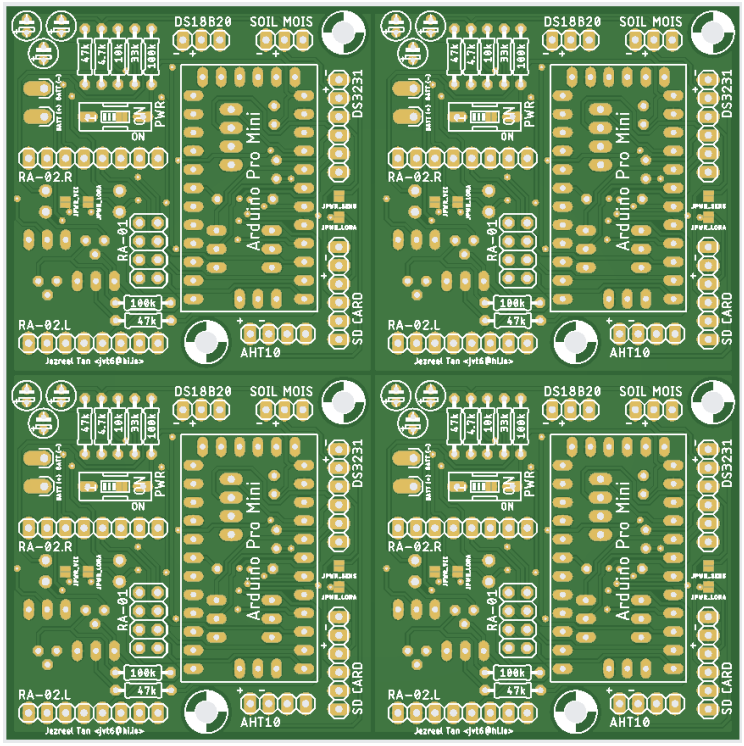


Figure 3.5: Assembled sensor node PCB before final enclosure installation.

System Assembly

To ensure reliable operation under the harsh environmental conditions in Iceland, all sensor components were enclosed in a weatherproof and waterproof housing.

The components were assembled neatly inside the enclosure to protect them from moisture, rain, snow, and temperature variations. Although the initial goal was to minimize the size of the sensor module—ideally approaching the size of the main PCB—the batteries were the primary limiting factor in reducing the overall form factor. Nevertheless, the final design remained compact, easy to handle, and capable of housing sufficient battery capacity for long-term operation.

Two enclosure designs were developed and evaluated to determine which configuration provided better performance and practicality. The first design integrated the LoRa antenna directly into the main sensor enclosure. This resulted in a more compact and mechanically robust module with fewer loose components. However, this configuration limited flexibility during installation, particularly in confined or uneven mounting locations.

The second design utilized an external waterproof enclosure connected via a cable, housing both the LoRa antenna and the AHT10 air temperature and humidity sensor. This configuration allowed the air sensor to be positioned above ground, resulting in more representative air temperature and humidity measurements. However, this design increased exposure of the sensor and cabling to environmental elements, requiring additional protection to prevent potential damage.

After assembly, all seams, cable entry points, and enclosure openings were sealed using silicone sealant to ensure watertightness. This final sealing step was critical in protecting the electronics from water ingress caused by rain, snow, and meltwater, thereby improving the overall durability and reliability of the system.

3.3.2. Base Station Module

The base station module consists of a low-power microcontroller unit (MCU) and a LoRa module. The MCU connects directly to the internet via Wi-Fi to synchronize time and is powered through a mains outlet to ensure a stable and continuous power supply. Using its internal clock, the MCU enters sleep mode between intervals and periodically checks the time through the internet to ensure accurate wake-up timing.

At scheduled intervals, the base station MCU sends a signal to all sensor nodes to initiate data collection. At this time, all sensor nodes are expected to be awake and ready to transmit data. Once the base station receives all required data, it broadcasts the current date and time to ensure that all sensor nodes remain time-synchronized. This process allows the sensor nodes to be synchronized on an hourly basis.

After synchronization, the base station uploads the collected data to the cloud, where it is stored and made available for viewing and further data analysis by researchers.

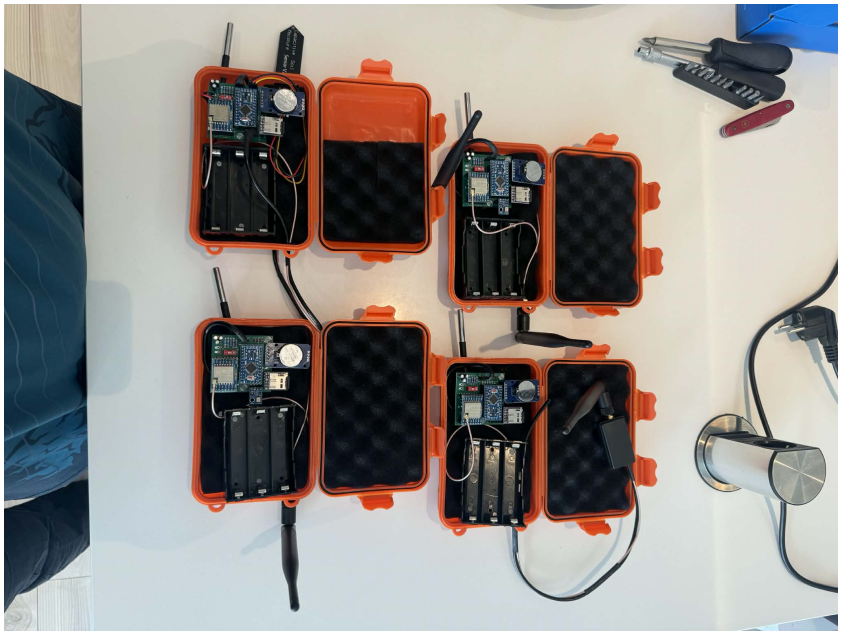


Figure 3.6: Weatherproof enclosure design for the buried sensor node and external sensor arrangement.

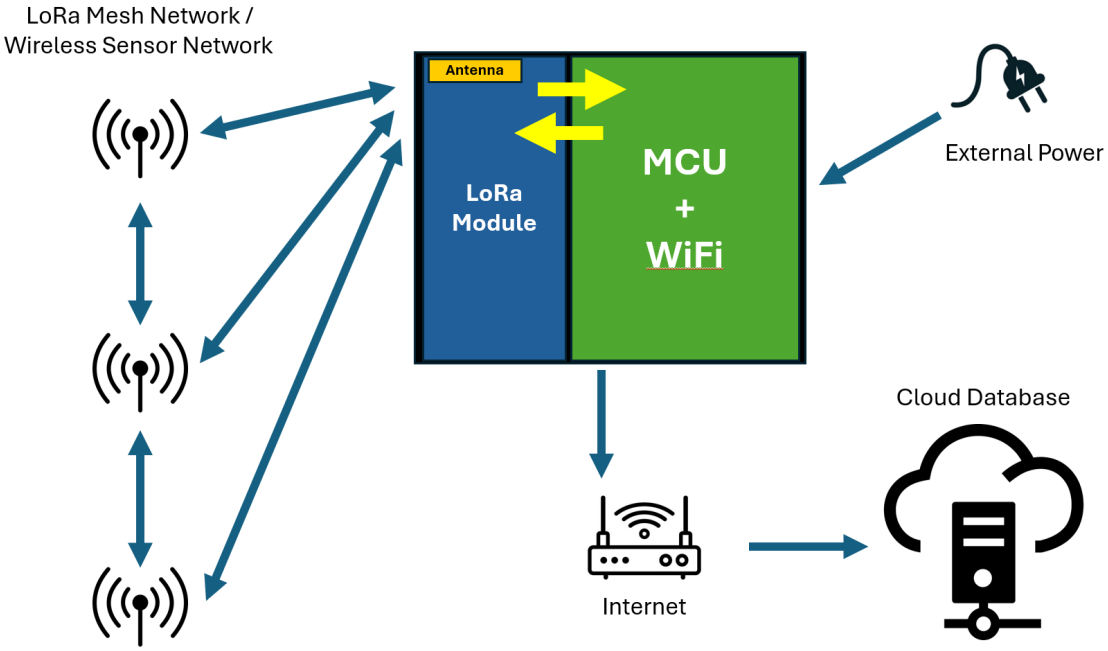


Figure 3.7: Base station module with LoRa transceiver and Wi-Fi connectivity for cloud upload.

Device Components

LoRa Module (Ra-02 SX1278 433Mhz)

The base station uses the same Ra-02 LoRa module as the in-situ sensor nodes to maintain consistency across the system. Using the same LoRa module simplifies system design, firmware development, and communication compatibility between the base station and sensor nodes. Additionally, this approach reduces component variation and ensures reliable and uniform communication performance throughout the network.

Microcontroller (LOLIN D1 mini)

Initially, during the concept design and planning stage, a Raspberry Pi was considered as the primary choice for the base station. Based on a review of similar theses and related projects, many researchers used a Raspberry Pi for comparable applications. However, after further development and evaluation, it was concluded that a Raspberry Pi would be excessive for the requirements of this system. In addition, it is relatively costly, and several lower-power microcontroller-based alternatives were found to be more suitable for the application.

Eventually, the LOLIN D1 mini was selected as the base station controller. The board is based on the ESP8266EX chip, which provides built-in Wi-Fi connectivity. It supports SPI communication, which is required for interfacing with the LoRa module, and operates at 3.3 V with an 80 MHz clock frequency and 4 MB of flash memory. These specifications were sufficient for the data handling, communication, and synchronization tasks of the base station. The board can be programmed using C/C++, allowing efficient use of its limited flash memory and providing flexibility for firmware customization. Additionally, the module can be easily powered through a USB Type-C connection, enabling the use of common power adapters for continuous operation.

System Assembly

The base station module has a relatively simple hardware configuration, which made assembly and integration straightforward. Since both the microcontroller and the LoRa module operate at 3.3 V, they can be directly interconnected without the need for additional level-shifting circuitry. The SPI communication lines were connected

between the MCU and the LoRa module, and the antenna was securely attached to the LoRa module to ensure proper radio performance.

Power was supplied to the base station through a mains-powered adapter, allowing continuous operation without power constraints. After assembly, the microcontroller was programmed with the base station firmware, and basic functionality was verified. Proper antenna installation was ensured to avoid signal degradation during operation.

3.4. Firmware

The firmware for the proposed LoRa mesh wireless sensor network was developed in C/C++ using an object-oriented programming (OOP) approach. The system is based on the ATmega328P microcontroller, which is well-supported within the Arduino development ecosystem and allows efficient low-level hardware control. An object-oriented structure was selected to improve modularity, scalability, and maintainability of the system. By encapsulating functionality into separate classes (e.g., sensor handling, LoRa communication, data management, and power control), the firmware becomes easier to debug, extend, and adapt for future deployments or additional sensing requirements.

Firmware development and flashing were performed using the Arduino IDE. Arduino was selected because of its open-source architecture, comprehensive library support, and strong developer community, which streamline firmware development, enable efficient debugging, and provide flexibility for future system expansion. Standard Arduino libraries were utilized where appropriate, including libraries for SPI communication (LoRa module), I2C communication (environmental sensors), SD card interfacing, and low-power sleep control. Custom logic was implemented to manage mesh routing, data formatting, packet handling, and power optimization.

Two primary firmware implementations were developed:

1. **In Situ Sensor Node Firmware** – responsible for sensor data acquisition, packet formatting, mesh communication, local data backup, and low-power operation.

2. **Base Station Firmware** – responsible for receiving and decoding mesh packets, managing acknowledgments, and uploading validated data to the cloud platform.

The full source code repository is publicly available at:
`https://github.com/JezT25/turfroof_lora`

3.4.1. The LoRa Mesh System

During the early stages of algorithm development, particular attention was given to how sensor data could be distributed evenly, reliably, and robustly across the mesh network. The primary objective was not only to ensure that the base station received all measurements, but also to enhance network resilience by reducing dependence on a single communication path.

An architectural concept inspired by distributed ledger systems, such as blockchain, influenced the final design. In blockchain systems, a shared ledger acts as a synchronized and verifiable record of transactions across multiple participants, ensuring consistency and integrity of data. While the proposed system does not implement blockchain or cryptographic consensus mechanisms, it adopts a simplified distributed-ledger-inspired approach to improve robustness within the LoRa mesh network.

In this implementation, the base station initializes a structured data ledger template, which is propagated throughout the network. Each sensor node appends its most recent measurement to the ledger before forwarding it to neighboring nodes. Through multi-hop transmission, the ledger circulates across the mesh until all participating nodes have contributed their respective data entries.

This approach ensures that:

- Data redundancy is introduced at the network level.
- Nodes can verify whether their measurements have been successfully incorporated.
- The system reduces single points of failure.
- Network robustness is improved in the event of intermittent connectivity.

By allowing all nodes to maintain awareness of the aggregated dataset before final transmission to the base station, the system achieves greater reliability compared to a traditional star topology where only the base station holds the complete dataset. The final validated ledger is then transmitted to the base station for cloud upload and long-term storage.

Architecture

To simplify debugging and system validation during development, all LoRa data packets were transmitted as human-readable plaintext strings. Although this approach is less efficient than using compact binary packet structures, it significantly improved transparency during early testing and troubleshooting. Being able to directly inspect transmitted data without additional decoding tools allowed for faster identification of routing, formatting, and communication errors within the mesh network.

It should be noted that, for a production-level implementation, a packed data structure (e.g., a structured binary payload containing predefined variables) would be preferable. Such an approach would reduce payload size, improve transmission efficiency, and optimize airtime usage — particularly important in low-bandwidth LoRa systems.

The decision to retain plaintext transmission in the final prototype was primarily influenced by the system's early testing phase, where plaintext communication was initially implemented and validated. Maintaining this format ensured consistency and stability throughout development. Nevertheless, future iterations of the system would benefit from migrating to a structured binary packet format for improved scalability and performance.

Packet Structure

To initiate communication within the mesh network, one node — typically the base station — broadcasts a structured ledger packet. This ledger acts as a container for aggregated data and is circulated among all participating nodes. The packet consists of three main components: a header, reserved data fields for each node, and a checksum value used for basic integrity verification.

Each node in the network is assigned a unique hardware ID. This ID determines the fixed position within the packet where the node inserts its data. For example, if a node has a hardware ID of 3, it writes its measurement into the fourth allocated data field within the ledger. This positional mapping prevents data overwriting and ensures deterministic placement of measurements, allowing every node to know exactly where its contribution belongs within the packet.

Because all data is transmitted in plaintext format, each node is allocated a fixed number of characters. While a packed binary structure with defined variable types would have been more memory-efficient and flexible, fixed-width plaintext fields were chosen to simplify debugging, inspection, and demonstration of the system.

Each numerical value is formatted to support a range from –99.9999 to 999.9999. This range comfortably accommodates all measured parameters (e.g., temperature, humidity, soil moisture, battery voltage), ensuring no overflow occurs within the defined field size.

To maintain low computational overhead on the ATmega328P microcontroller, the checksum mechanism was intentionally simplified. Instead of implementing a standard CRC16 or similar algorithm, the checksum is computed as the arithmetic sum of numerical values from node 0 through node 7. While this does not provide the robustness of industry-standard CRC implementations, it offers a lightweight integrity check suitable for the processing and memory constraints of the hardware.

All data entries are comma-separated and enclosed within brackets ([]), forming a predictable and easily parseable structure. The packet header defines the type of data being transmitted, such as air temperature, relative humidity, soil temperature, soil moisture, battery voltage, or time/date synchronization.



Figure 3.8: Ledger packet format used for distributed data collection in the LoRa mesh network.

Time/date packets follow a slightly modified format. Instead of allocating separate fields per node, the packet's data fields are used collectively to distribute synchronized timestamp information across the network. This mechanism ensures that all nodes maintain consistent system time for logging and coordination purposes.



Figure 3.9: Time synchronization packet format used to align sleeping sensor nodes.

In total, each packet occupies 97 characters (97 bytes). Although plaintext transmission simplifies inspection, it inherently lacks confidentiality. To provide a basic level of security, packets are encrypted prior to transmission using the RC4 stream cipher. RC4 was selected due to its lightweight computational requirements and minimal memory footprint, making it suitable for the ATmega328P's limited RAM and processing capability.

An initial implementation using AES encryption was considered; however, it proved too computationally intensive for the microcontroller. AES encryption introduced noticeable delays in packet preparation, which negatively impacted mesh responsiveness and routing performance. Given the constrained resources of the ATmega328P, RC4 provided a practical compromise between security and system efficiency.

RC4 Encryption

The implemented function follows the standard RC4 structure, consisting of a key scheduling phase and a pseudo-random generation phase. A state array is initialized and permuted using the predefined ENCRYPTION_KEY. During encryption, a pseudorandom byte (rnd) is generated and XORed directly with each byte of the data buffer. Because RC4 is symmetric, the same function is used for both encryption and decryption. To conserve RAM, the implementation limits the state array to 255 bytes instead of the conventional 256, reducing memory consumption while maintaining functional encryption for this prototype. Although RC4 is not considered secure for high-security applications due to known vulnerabilities, it provides sufficient packet obfuscation for this research deployment where low power consumption and computational efficiency are prioritized.

The implemented function is shown below:

```
void LORA_MODULE_class::rc4EncryptDecrypt(char *data, uint8_t len)
{
    // Using RC4 because originally AES was the plan but the RAM/CPU on this thing cant take it.
    // RC4 uses 256 but this code is limited to 255. (uint8_t[255] = 256 bytes | uint16_t[256] = 512 bytes)
    // Save the RAM!

    uint8_t S[RC4_BYTES];
    for (uint8_t i = 0; i < RC4_BYTES; i++)
    {
        S[i] = i;
    }

    uint8_t j = 0, temp;
    uint8_t enc_len = strlen(ENCRYPTION_KEY);
    for (uint8_t i = 0; i < RC4_BYTES; i++)
    {
        j = (j + S[i] + ENCRYPTION_KEY[i % enc_len]) % RC4_BYTES;
        temp = S[i];
        S[i] = S[j];
        S[j] = temp;
    }

    uint8_t rnd = 0, i = 0; j = 0;
    for (uint8_t n = 0; n < len - 1; n++)   // -1 dont include null terminator
    {
        i = (i + 1) % RC4_BYTES;
        j = (j + S[i]) % RC4_BYTES;
        temp = S[i];
        S[i] = S[j];
        S[j] = temp;
        rnd = S[(S[i] + S[j]) % RC4_BYTES];
        data[n] ^= rnd;
    }
}
```

Figure 3.10: RC4-based packet encryption routine implemented for lightweight mesh communication security.

The encrypted packet structure preserves low computational overhead while still providing basic obfuscation for plaintext LoRa transmissions during early development and testing.

Mesh Network

Once a ledger is transmitted, any node within range can receive it, provided that it is properly formatted. Upon reception, the node decrypts the packet, extracts the data, and compares it with its local database. The node then updates its local database using the data from the received ledger and updates the ledger using its own stored data. After updating, the ledger may be relayed to other nodes to allow them to update their respective databases. This process continues until all nodes possess a fully updated ledger.

Relaying is not automatically performed for every received packet. If unconditional relaying were implemented, the mesh network would fall into an infinite rebroadcast loop. Before relaying a ledger, the node must determine whether retransmission is necessary. If a node receives a ledger that differs from its local database, it indicates that at least one node in the network is missing data. In this case, the receiving node updates the ledger using its local database and relays it, ensuring that the node which originally transmitted the incomplete ledger eventually receives the updated information.

For example, if Node 3 receives a ledger that does not contain data from Node 3, it can infer that somewhere in the network there exists a node that does not have Node 3's data. Node 3 then updates the ledger and relays it. This mechanism is not limited to the node's own data. If Node 3 receives a ledger missing Node 5's data, but Node 3 already has Node 5's data stored locally from a previous transmission, Node 3 can update the ledger and retransmit it. This eliminates the need for the data to travel multiple hops back to Node 5 before being redistributed, improving overall efficiency.

The only situation in which a node does not relay a ledger is when the received ledger is identical to its local database. In this case, all participating nodes already possess the same data, and retransmission would only create unnecessary traffic and potential infinite looping. Therefore, the packet is ignored.

In a linear topology scenario, the base station may only be within range of Node 1; Node 1 within range of the base station and Node 2; Node 2 within range of Node 1 and Node 3; and so forth. When the base station sends a packet, Node 1 receives it, updates the ledger, and retransmits it to nodes within range. This creates a chain reaction until the last node updates the ledger. The fully updated ledger then

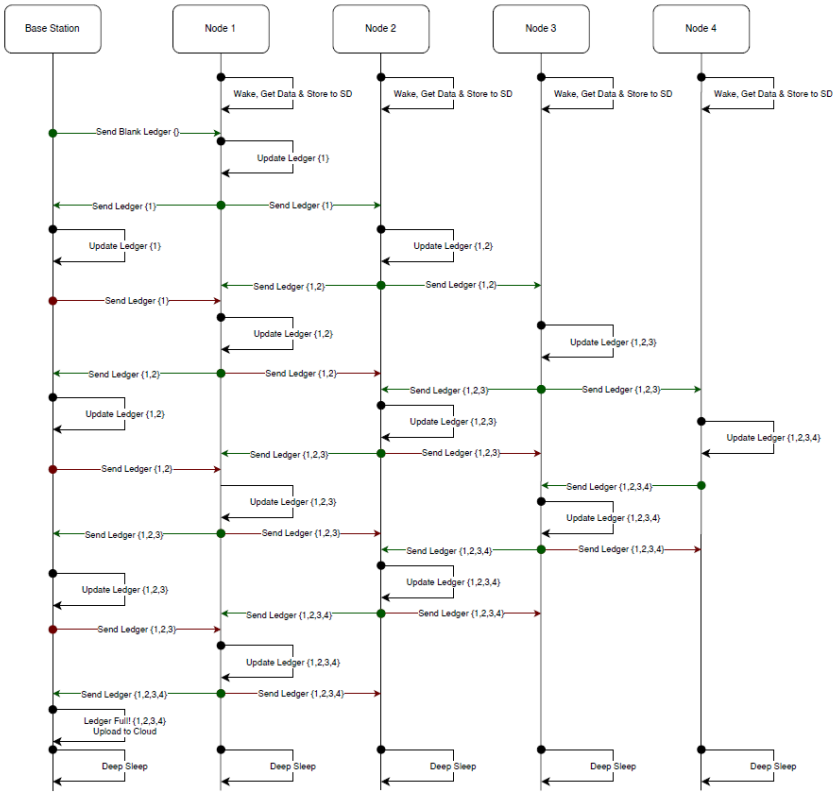


Figure 3.11: Relay and update process used to propagate missing sensor data through the mesh network.

propagates back toward the base station through another chain reaction. Once the base station receives the complete ledger, the data is saved and uploaded to the cloud.

Collision Avoidance (CSMA/CA)

To prevent simultaneous transmissions that may cause packet collisions, Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) is implemented. Since the LoRa module cannot transmit and receive at the same time, collision avoidance is necessary to ensure reliable communication.

Before transmitting, a node first checks the LoRa channel noise level. If the channel is busy, indicating that packets are currently being transmitted, the node waits until the channel becomes idle before attempting to send. This reduces the probability of collision.

However, a node cannot wait indefinitely. Therefore, a timeout mechanism is implemented. If all nodes were assigned identical timeout values, collisions could still occur when timeouts expire simultaneously. To avoid this, each node multiplies the base timeout value by its hardware ID. This ensures that each node has a different timeout duration. In the worst-case scenario where all nodes intend to transmit and reach their timeout, transmissions will occur at different times due to the hardware ID multiplier.

Additionally, if a node receives a valid packet while waiting to transmit, it reevaluates the received packet, updates or discards it accordingly, and restarts its timeout counter. This effectively delays its transmission and reduces the likelihood of congestion. This mechanism ensures that channel noise decreases before additional packets are introduced.

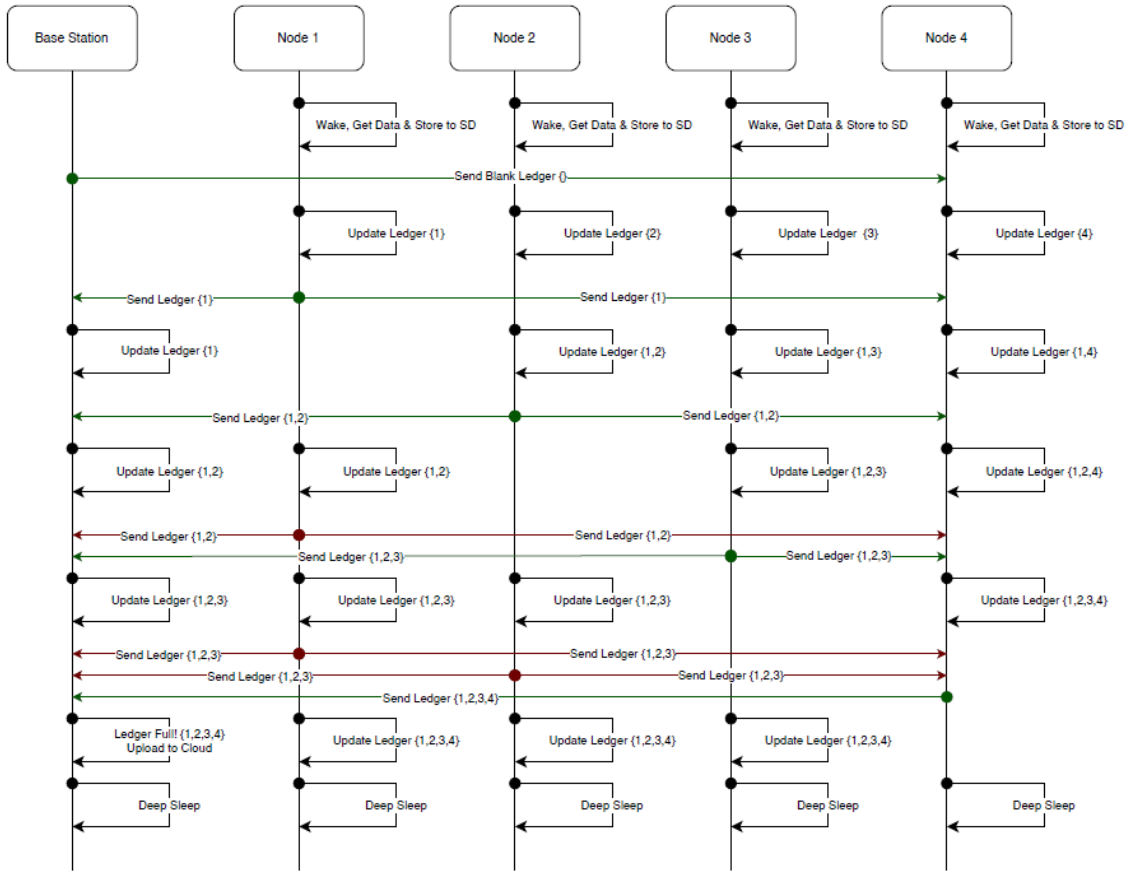


Figure 3.12: CSMA/CA timing strategy used to reduce collisions during mesh packet transmission.

In a scenario where all nodes are within range of each other, meaning a single transmission is received by all nodes, those with lower hardware IDs transmit first due to shorter timeouts. For example, once all nodes update their local databases,

Node 1 (with the lowest ID) transmits first. Upon receiving Node 1's transmission, all other nodes stop their pending transmissions, update their databases, and restart their timeout process. This continues sequentially until all nodes have shared their data and the base station receives the complete dataset for upload to the cloud.

Although this collision avoidance mechanism significantly reduces packet collisions, it is not perfect. There may be rare timing conditions where two nodes synchronize unintentionally and transmit simultaneously. However, due to the mesh structure of the network, the loss of a single packet is recoverable. Since data is continuously relayed among nodes, missed information can be recovered from neighboring nodes in subsequent transmissions.

3.4.2. LoRa Firmware Configuration

TODO: Add details about the selected radio parameters and legal operating settings.

```
// CSMA/CA Settings
#define SEND_ATTEMPTS       2
#define BACKOFF_MUL         37
#define CSMA_NOISE_LIM      -90
#define CSMA_TOUT_MUL       250
#define CSMA_TOUT_MIN       1500
```

Figure 3.13: LoRa radio configuration settings selected for legal and reliable communication.

The configuration was chosen to balance range, airtime, and reliability while remaining within practical low-power LoRa operating limits for the deployed system.

3.4.3. In Situ Sensor Behavior and Algorithm

The in situ node operates in a low-power duty cycle. It enters deep sleep mode, wakes up using an RTC timer, gathers sensor data, transmits data to the mesh network, and then returns to deep sleep. This cycle repeats continuously throughout deployment.

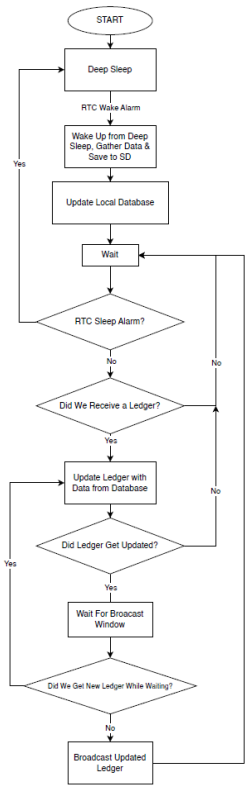


Figure 3.14: Operational duty-cycle diagram of the in-situ sensor node during deployment.

Sensor Data Reading

The in situ node contains multiple sensors as previously described: air temperature and humidity, soil temperature, soil moisture, and battery voltage. These sensors communicate using different interfaces including I2C, 1-Wire serial, and analog-to-digital conversion (ADC).

Air Temperature and Humidity (AHT10)

The AHT10 sensor communicates using the I2C protocol. The microcontroller sends a measurement command to the sensor, after which the sensor internally performs the necessary signal processing and calibration. The processed temperature and humidity data are then transmitted digitally over the I2C bus to the microcontroller. This simplifies firmware implementation, as the sensor provides calibrated digital output directly.

Soil Temperature

The DS18B20 soil temperature sensor communicates using the 1-Wire protocol. Each sensor contains a unique address, allowing individual identification if multiple sensors share the same bus. The microcontroller initiates a temperature conversion command and reads the digital temperature value directly from the sensor, eliminating the need for analog calibration.

Soil Moisture

The soil moisture sensor operates on a capacitive principle. Using a comparator-based circuit (**TODO:** Confirm the comparator chip model), the dielectric constant of the soil affects the sensor output voltage.

With a 3.3 V supply input, the sensor outputs an analog voltage proportional to soil moisture. This voltage is read using the 10-bit ADC of the ATmega328P.

ADC resolution:

$$\text{Resolution} = \frac{V_{ref}}{2^{10}} = \frac{3.3V}{1024} \tag{3.1}$$

The raw ADC value is converted to soil moisture percentage using calibration data obtained through the gravimetric method (see Chapter 4).

Battery Voltage

The battery voltage is measured using a voltage divider consisting of:

- $R_1 = 10\ k\Omega$
- $R_2 = 33\ k\Omega$

Voltage divider formula:

$$V_{out} = V_{battery} \times \frac{R_2}{R_1 + R_2} \tag{3.2}$$

Rearranged to calculate battery voltage:

$$V_{battery} = V_{ADC} \times \frac{R_1 + R_2}{R_2} \tag{3.3}$$

The ADC is 10-bit resolution. The digital value is converted to voltage using the reference voltage and the equation above.

3.4.4. Base Station Module Program

The base station module operates similarly in low-power mode. It enters deep sleep and periodically wakes to query the global time from the internet using pool.ntp.org. When scheduled, it transmits a blank ledger to initiate mesh data collection. It then waits until it receives a fully updated ledger containing data from all nodes. Once all required data has been gathered, it uploads the information to the cloud.

After successful data collection, the base station synchronizes all nodes by sending a formatted packet containing the correct time obtained from pool.ntp.org. Although nodes are equipped with RTC modules, periodic synchronization ensures that all nodes remain aligned and wake simultaneously, preventing missed data requests.

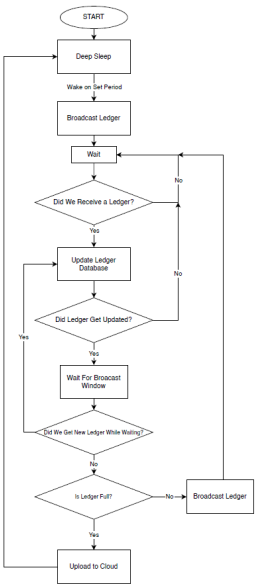


Figure 3.15: Base station data acquisition and cloud upload workflow for the deployed network.

Uploading the Data

The cloud platform selected for this project is ThingSpeak. To upload data, the base station performs an HTTP request using the ThingSpeak API key. **VERIFY:** Confirm whether the request uses POST or GET and add the returned HTTP status code.

Since the system can support up to 8 nodes, the base station uploads each node's data individually using their respective API keys.

3.5. Cloud

A cloud-based platform was required to allow researchers to access sensor data remotely, as the turf houses are located far from the research facilities and frequent on-site visits are impractical. Storing data in the cloud enables continuous monitoring and simplifies data access, making it easier for researchers to view and analyze measurements without physically retrieving the devices.

Several criteria were considered when selecting a suitable cloud platform. These included reliability, accessibility, low cost, ease of integration with embedded systems, and the ability to store and visualize time-series sensor data. While more advanced platforms such as AWS or Microsoft Azure provide robust infrastructure, they were considered unnecessarily complex and costly for the scope of this project.

Based on these criteria, ThingSpeak was selected as the cloud platform. ThingSpeak provides reliable data storage and built-in visualization tools that are well suited for IoT applications. Given the expected data transmission rate of the system, the required storage and upload frequency fall within the limits of the free service tier, making it a cost-effective solution. In this implementation, sensor data uploaded to the cloud is stored and made readily accessible for remote viewing and further analysis by researchers.

3.6. Testing

During the development of the project, multiple tests were conducted to evaluate system performance and reliability. These tests included power consumption measurements, LoRa range testing, and several preliminary tests carried out at Háskóli Íslands prior to the final deployment. The testing phase provided valuable insights

into early system limitations and failures, allowing necessary modifications and improvements to be made before final deployment.

3.6.1. In Situ Sensor Node Power Consumption Tests

After the initial assembly of a functional sensor node, power consumption testing was conducted using a precision laboratory power supply (Rohde & Schwarz HMP2030), capable of measuring current down to 100 µA. The sensor module was powered at 4.2 V to represent a fully charged battery condition, and the power supply's current readout was used to observe and record current consumption.

During testing, the sensor node was programmed to cycle through multiple operating states in a continuous loop. Each state was maintained for 10 seconds before transitioning to the next state, allowing the current readings to stabilize. State transitions were controlled entirely through firmware to ensure consistent and repeatable behavior. The measurement cycle was allowed to run continuously, and power consumption was observed over three complete loops to verify repeatability.

The operating states tested included the following:

- **Deep sleep mode**
- **Awake mode with sensors enabled**
- **LoRa transmission and reception mode**
- **Half-sleep mode**, in which the sensor modules were disconnected while the LoRa module remained fully powered and the MCU in a light sleep mode.

Deep Sleep Mode:

Deep sleep mode completely shuts down most of the node's hardware components to minimize power consumption for long-term operation. In this mode, the system disables communication interfaces such as SPI and I²C, turns off GPIO modules and the LoRa radio, and sets all pins to a low-power state. Interrupts from the LoRa module are removed, and only the real-time clock (RTC) interrupt is enabled to wake the device at scheduled intervals. The microcontroller then enters the gotosleep() state where almost all processes are halted. When the RTC interrupt occurs, the system wakes up, re-enables the LoRa interrupt, powers the modules back on, and reinitializes the LoRa configuration before resuming normal operation. This mode

is designed to achieve maximum energy savings for battery-powered wireless sensor nodes that operate for long periods between transmissions.

Light Sleep Mode:

Light sleep mode reduces power consumption while still keeping certain communication capabilities active, particularly the LoRa module for receiving data. In this mode, only non-essential devices and GPIO modules are powered down, while LoRa is kept in receive mode so the node can wake immediately when a packet arrives. The I²C communication is disabled and unused pins are set to low-power states, but the system maintains the LoRa interrupt to trigger wake-up. When the interrupt from the LoRa module is detected, the microcontroller exits the sleep state and resumes execution. Compared to deep sleep, light sleep consumes more power but allows faster response times and continuous listening for incoming LoRa messages, making it suitable for nodes that must remain partially active within the wireless sensor network.

3.6.2. Háskóli Íslands Preliminary Test I

After all system components had been assembled and verified, an initial set of local preliminary tests was conducted at Háskóli Íslands. With permission from **TODO:** Add the name of the approving person and Dorotea, testing was carried out on the turf roofs located within the university campus.

A single LoRa base station was installed in Helgi's office, where it was connected to a computer configured for continuous monitoring of incoming data via the cloud platform. This setup allowed real-time observation of device status, packet reception, and data integrity throughout the test period.

Four sensor devices were deployed across the surrounding area, primarily in the vicinity of the VR-II and VR-III buildings. All devices were positioned within an approximate radius of 100 meters from the base station, ensuring stable connectivity while enabling early validation of system reliability under realistic deployment conditions. For this test, each sensor was powered by a single battery cell in order to evaluate power consumption and determine the operational lifetime achievable with one cell.

The preliminary test phase was conducted over a three-week period, from **June 8 to June 29, 2025**. During this time, the system operated continuously, providing insight into long-term stability, data consistency, and potential environmental effects on communication performance.

3.6.3. LoRa Range Tests

Based on insights gained from the initial field tests, it became necessary to conduct more extensive range testing to further evaluate system performance under different conditions. Multiple LoRa configurations were evaluated to determine the most optimal balance between communication range and performance. Previous studies have demonstrated that LoRa is capable of achieving significant transmission distances under suitable conditions, and these claims needed to be validated using the current hardware setup and deployment environment.

The LoRa transceiver supports a wide range of configurable parameters, including operating frequency, transmission power, bandwidth, sync word, spreading factor, coding rate, and preamble length. For these experiments, several of these parameters were kept constant—namely frequency, transmission power, sync word, coding rate, and preamble length—in order to ensure fair and repeatable comparisons between tests. The parameters that were primarily adjusted during testing were bandwidth and spreading factor, as these have the most direct and significant impact on range, data rate, and signal robustness.

Key LoRa configuration parameters:

- **Bandwidth (BW):** Determines the width of the frequency band used for transmission. Lower bandwidth improves receiver sensitivity and increases range, but reduces data rate.
- **Spreading Factor (SF):** Controls how many chirps are used to encode each symbol. Higher spreading factors increase range and resistance to noise at the cost of lower data rates and longer transmission times.
- **Transmission Power:** Defines the output power of the radio. This was kept constant to comply with regulatory limits and to isolate the effects of BW and SF.
- **Coding Rate (CR):** Adds forward error correction to improve data reliability. This was fixed to maintain consistent error correction across tests.
- **Preamble Length:** Used for receiver synchronization. A fixed preamble ensured consistent packet detection behavior.

Bandwidth and spreading factor were the primary variables during testing because together they directly influence link budget and receiver sensitivity. Adjusting other parameters would either violate regulatory constraints or introduce additional variables that make performance comparisons less reliable. By focusing on BW and

SF, it was possible to observe clear trade-offs between range, signal quality, and data throughput.

The range tests were conducted in Vellir, located in Hafnarfjörður. The base station was mounted on a fence at approximately chest height to simulate a realistic low-elevation deployment. A mobile sensor node was placed inside a vehicle and connected to a laptop for real-time monitoring.

During the tests, the vehicle was driven around the surrounding area while the base station transmitted randomly encoded data packets once per second. The received signal strength indicator (RSSI) values were recorded continuously, and received packets were inspected to verify data integrity and detect any corruption. This approach allowed both communication reliability and signal strength to be evaluated simultaneously under real-world conditions.

3.6.4. Háskóli Íslands Preliminary Test II

Building on the insights gained from the previous preliminary tests and subsequent system improvements, a longer-duration local test was conducted at Háskóli Íslands. Due to seasonal conditions approaching winter, access to the turf roofs was no longer permitted, as they were required for ongoing studies by the responsible staff member.

As a result, four sensor devices were redeployed across the campus within an approximate radius of 100 meters from the base station, using the same monitoring and data collection setup as in the earlier tests. The base station remained connected to a dedicated computer for continuous cloud-based data monitoring. In this test, each sensor was powered by a single battery cell, consistent with the previous test configuration.

For this test phase, the antenna system was upgraded to improve communication performance. Larger and higher-quality antennas were installed on the sensor nodes, and an outdoor antenna was mounted outside Helgi's office to enhance base station reception and transmission.

This second preliminary test was conducted over an extended period, from **August 1 to October 6, 2025**, allowing for evaluation of long-term system stability, communication reliability, and the impact of environmental conditions on performance.

3.7. Deployment in Keldur

After incorporating lessons learned from the previous test phases, the system was considered ready for full deployment. In collaboration with Dorotea, a deployment visit was arranged to **Keldur**. **TODO:** Add the final Keldur site information.

For this deployment, each sensor node was configured with the maximum number of battery cells supported by the enclosure, which was three cells per sensor. All sensors were fully sealed and prepared for long-term outdoor operation prior to deployment.

Upon arrival at Keldur, the network infrastructure was already in place from earlier testing. The Wi-Fi router was located in the attic of one of the turf houses. The LoRa base station was installed at this location and connected to the existing network, allowing immediate communication with the cloud platform. Once verified, the base station equipment was discreetly concealed to avoid visual impact, as the site operates as a museum.

Following the base station setup, four sensor units were installed around the area, with one sensor assigned to each turf roof.

TODO: Add details describing sensor placement, mounting method, orientation, and deployment photographs.

After installation, all equipment was confirmed to be operational and securely concealed. The long-term deployment test commenced on **October 17, 2025**, and is currently ongoing. **VERIFY:** Add the end date when the deployment is complete.

3.8. Calibration Method

To determine the accuracy of the off-the-shelf soil moisture sensors relative to an industry-standard instrument, the sensors were compared side by side with a HOBO MX230x Soil Moisture and Temperature Data Logger. The HOBO device is a fully integrated soil moisture logger with Bluetooth-based wireless data offload capability and measures both soil moisture and temperature.

To ensure scientific accuracy, the gravimetric method was performed alongside the sensor comparison to establish the actual soil moisture content. Calibration was conducted at **TODO:** Add the laboratory name located at VR-3.

An Icelandic soil sample was collected and oven-dried at 105 °C for 32 hours (from 11:00 AM to 8:00 PM the following day) to remove all moisture and establish a dry baseline condition. After drying, the soil was allowed to cool to room temperature before testing. A measured portion of the dried soil was then placed in a container, and all sensors were installed side by side to ensure identical environmental conditions.

Water was added incrementally in 10% increments relative to the dry soil weight. A spray mister was used to distribute water evenly throughout the soil. After each increment, sufficient time was allowed for uniform saturation before measurements were recorded. At each moisture level, multiple readings were collected from all sensors to ensure consistency and reduce variability. This procedure was repeated for each incremental moisture level.

The HOBO sensor provides measurements of volumetric water content (VWC) in mineral soils, with a specified measurement range of 0.00 to 0.64 m^3/m^3. In this experiment, mineral soil configuration was used. Although the device also allows alternative output settings, the selected configuration was kept consistent throughout the experiment. Where necessary, conversion equations were applied during data analysis to ensure comparability of results.

The experiment continued until the soil reached full saturation and additional water resulted in surface pooling, indicating that no further absorption could occur. At this point, the calibration process was concluded and all measurements were recorded and stored for analysis.

To ensure reliability and repeatability, the entire calibration procedure was conducted twice. The first experiment was performed on November 8, and the second on December 16.

3.8.1 Reference Instrument: HOBO MX230x Soil Moisture and Temperature Data Logger

The HOBO MX230x Soil Moisture and Temperature Data Logger was used as the reference instrument for calibration. This commercial device is widely used in environmental monitoring studies and is also currently utilized by research teams working on Icelandic turf roof monitoring. Using this instrument as the reference ensures consistency and comparability with existing research efforts.

The soil moisture sensor measures volumetric water content (VWC) with a measurement range of 0.00 to 0.64 m^3/m^3 in mineral soils and 0.00 to 0.70 m^3/m^3 in soilless media. The typical accuracy is ± 0.031 m^3/m^3 in mineral soils (solution EC < 8 dS/m), with a resolution of 0.001 m^3/m^3. The sensor operates at a dielectric measurement frequency of 70 MHz.

The integrated temperature sensor (MX2307) has a measurement range of −40 °C to 100 °C with an accuracy of ±0.2 °C to ±0.25 °C depending on the temperature range, and a resolution of 0.04 °C. The logger supports adjustable logging intervals from 1 second to 18 hours and can store up to 110,000 measurements. The device uses Bluetooth Low Energy for wireless data retrieval and is powered by a 3.6 V lithium battery with a typical operational life of up to several years, depending on logging configuration.

The manufacturer also specifies that the soil moisture measurements can be post-calibrated using custom calibration equations if higher accuracy is required or if soil conditions differ from standard mineral soils. This feature further supports its suitability as a calibration reference instrument for this study.

4. Results and Discussion

4.1. Overview

This chapter presents the results obtained from laboratory power consumption testing, preliminary field trials at Háskóli Íslands, LoRa range evaluation, long-term deployment in Keldur, and soil moisture calibration. The results are analyzed in relation to the system requirements defined in Chapter 3, particularly long-term low-power operation, reliable buried communication, mesh robustness, and calibration accuracy.

4.2. Power Consumption Results

4.2.1. Packet Sending Time Calculation

The in-situ sensor node operates under a strict duty cycle constraint, remaining active for a maximum of 5 minutes per hour while spending the remaining 55 minutes in deep sleep. The transition from deep sleep to full operation, including sensor activation via MOSFET, data acquisition, and shutdown, takes approximately 3.1 seconds.

LoRa packet airtime is determined by the sum of preamble and payload durations:

$$T_{packet} = T_{preamble} + T_{payload} \tag{4.1}$$

With spreading factor $SF = 10$, bandwidth $BW = 125$ kHz, and preamble length $N_{preamble} = 8$, the symbol time is:

$$T_{sym} = \frac{2^{SF}}{BW} = \frac{2^{10}}{125000} = 0.008192 \text{ s} \tag{4.2}$$

The preamble duration is therefore:

$$T_{preamble} = (N_{preamble} + 4.25) \cdot T_{sym} \approx 0.1 \text{ s} \tag{4.3}$$

The payload duration depends on the payload size $PL$and is given by:

$$T_{payload} = N_{payload} \cdot T_{sym} \tag{4.4}$$

For the selected configuration (CRC enabled, explicit header, low data rate optimization, coding rate 4/5), this simplifies to:

$$N_{payload} = 8 + 5 \cdot \left\lceil \frac{8PL + 4}{32} \right\rceil \tag{4.5}$$

Thus, total packet airtime becomes:

$$T_{packet} = \left(20.25 + 5 \cdot \left\lceil \frac{8PL + 4}{32} \right\rceil\right) \cdot 0.008192 \tag{4.6}$$

For a 96-byte payload, the airtime is approximately 1.19 seconds per transmission.

Assuming six sensor data types (e.g., temperature, humidity, soil temperature, battery level, and timestamp) are transmitted eight times per hour, the total transmission time is:

$$(1.19 \text{ s} \times 8) \times 6 \approx 57 \text{ s} \tag{4.7}$$

This is approximated to 1 minute of transmission time per hour. Then we can assume that the rest is spent receiving LoRa packets and operating in standby mode, which is approximately 237 seconds.

4.2.2. Operating Modes

Table 4.1: Operating states, measured current consumption, and duration per hour.

Operating State	**Measured Current (mA)**	**Duration per Hour**	**Notes**
Deep Sleep	0.3 mA	55 minutes	All modules off, MCU in deep sleep
Awake (Sensors ON / LoRa OFF)	9.5 mA	3 s	Sensor initialization and data acquisition
LoRa RX/TX	119.5 mA	60 s	Transmission and reception of LoRa packets
Half Sleep	12.2 mA	237 s	Sensors off, MCU in low-power mode, LoRa in standby

The average current consumption is calculated as:

$$I_{avg} = \frac{\sum I_n t_n}{T} \tag{4.8}$$

where $I_n$is the current in each state, $t_n$is the duration of each state, and $T = 3600$ sis the total time per hour.

Substituting the measured values:

$$\begin{aligned} I_{avg} &= \frac{(0.3 \times 3300) + (9.5 \times 3) + (119.5 \times 60) + (12.2 \times 237)}{3600} \\ &= \frac{990 + 28.5 + 7170 + 2891.4}{3600} \\ &\approx 3.08\,\text{mA} \end{aligned} \tag{4.9}$$

Thus, the average current consumption of the sensor node under worst-case conditions is approximately **3.08 mA**.

Given a total battery capacity of 10,500 mAh (three 3,500 mAh cells in parallel), the estimated operational lifetime is:

$$\frac{10500}{3.08} \approx 3409 \text{ hours} \approx 142 \text{ days } (\sim 4.7 \text{ months}) \tag{4.10}$$

This estimate assumes ideal battery performance. In practice, environmental factors—particularly low temperatures in Iceland—will reduce effective battery capacity and shorten the operational lifetime.

For comparison, a single 3,500 mAh cell would yield an estimated lifetime of approximately:

$$\frac{3500}{3.08} \approx 1136 \text{ hours} \approx 47 \text{ days} \tag{4.11}$$

This highlights the importance of parallel battery configuration for extended deployment.

Overall, the achieved lifetime is sufficient for the intended application. Turf roofs are periodically inspected and maintained, allowing battery replacement to be integrated into routine maintenance cycles. This ensures continuous operation of the sensor nodes while supporting long-term monitoring of soil conditions and turf roof integrity.

4.3. Initial Field Deployment

An initial field test was conducted at various locations around the University of Iceland (Háskóli Íslands) campus. The primary objective of this deployment was not to evaluate the final performance of the system, but rather to validate the functionality of the preliminary hardware, firmware, and communication system under real outdoor conditions. Since turf roofs are present at several buildings on campus, including the turf-covered bicycle shed, these locations provided an accessible environment for testing while allowing rapid troubleshooting and hardware modifications during development.

Four prototype sensor nodes (HW_1, HW_3, HW_5, and HW_7) were deployed to collect air temperature, relative humidity, soil temperature, and soil moisture data. Measurements were transmitted through the LoRa network to the base station and uploaded to the cloud for analysis. Weather station data from the University of Iceland were used as a reference for comparison.

4.3.1. Air Temperature and Relative Humidity

Figure 4.1 compares the measured air temperature from the prototype sensor nodes with the nearby weather station, while Figure 4.2 presents the corresponding relative humidity measurements.

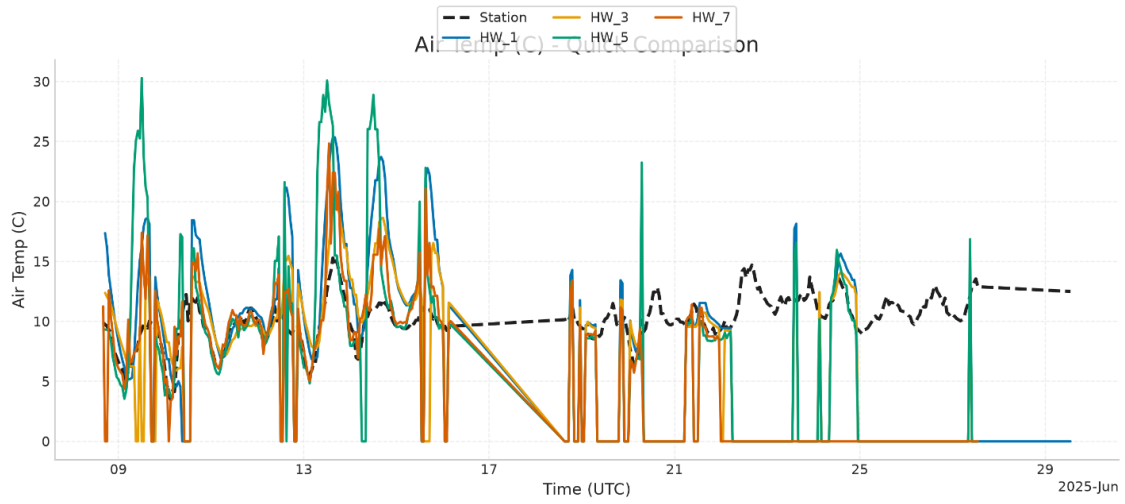


Figure 4.1: Air temperature measurements during the preliminary field deployment.

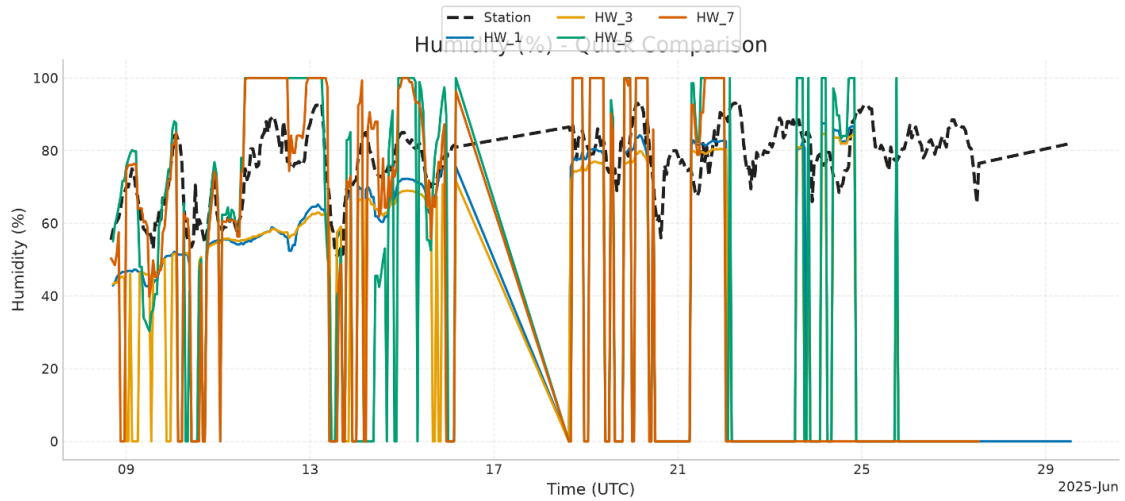


Figure 4.2: Relative humidity measurements during the preliminary field deployment.

The prototype nodes successfully collected environmental measurements throughout the deployment period and generally followed the overall trends observed by the weather station. However, numerous interruptions, missing samples, and unstable readings were observed. These inconsistencies were expected because the firmware and hardware were still under active development, with frequent updates and adjustments being performed throughout the testing period.

One of the major issues identified during this deployment was poor LoRa communication performance. During testing, the sensor nodes were initially equipped with commercially available antennas that were later found to be designed primarily for Wi-Fi frequencies rather than the **433 MHz LoRa frequency** used by the system. As a result, signal quality was significantly degraded, leading to frequent packet loss, communication interruptions, and incomplete datasets. This issue was identified during the preliminary deployment and later resolved by replacing the antennas with properly tuned 433 MHz LoRa antennas before the final deployment.

4.3.2. Soil Temperature and Soil Moisture

Figure 4.3 presents the measured soil temperature, while Figure 4.4 shows the corresponding soil moisture measurements collected during the preliminary deployment.

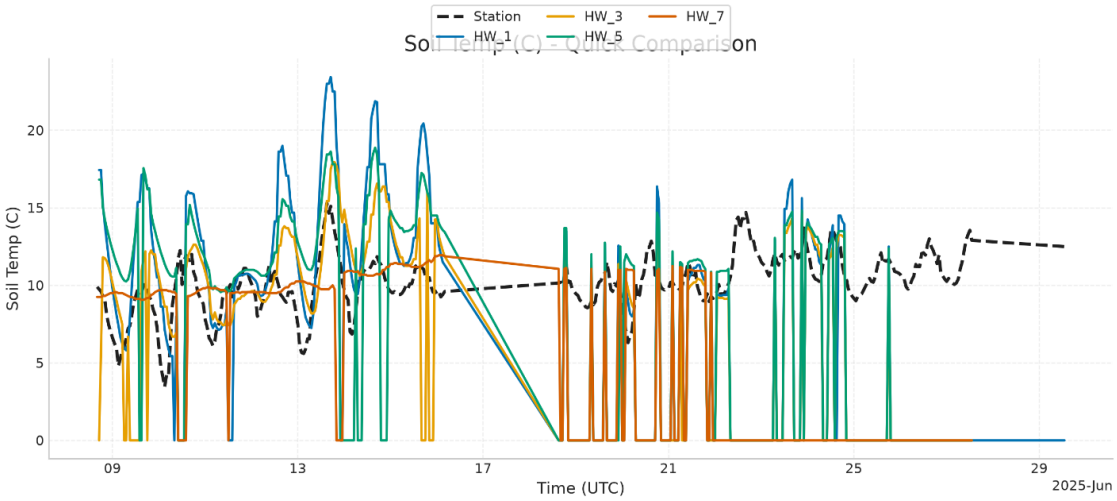


Figure 4.3: Soil temperature measurements during the preliminary field deployment.

The prototype sensor nodes successfully measured subsurface temperature and soil moisture while installed beneath the turf roofs. Similar to the later deployments, soil temperature exhibited considerably smaller fluctuations than ambient air temperature due to the thermal buffering properties of the surrounding soil.

The soil moisture measurements, however, showed considerable variability and several unrealistic values exceeding the expected physical range. At this stage of development, the moisture sensors had not yet undergone final calibration, and firmware updates affecting sensor scaling were still being tested. Combined with the communication issues caused by the incorrect antennas, the resulting datasets contained numerous missing samples and erroneous measurements. Nevertheless, the

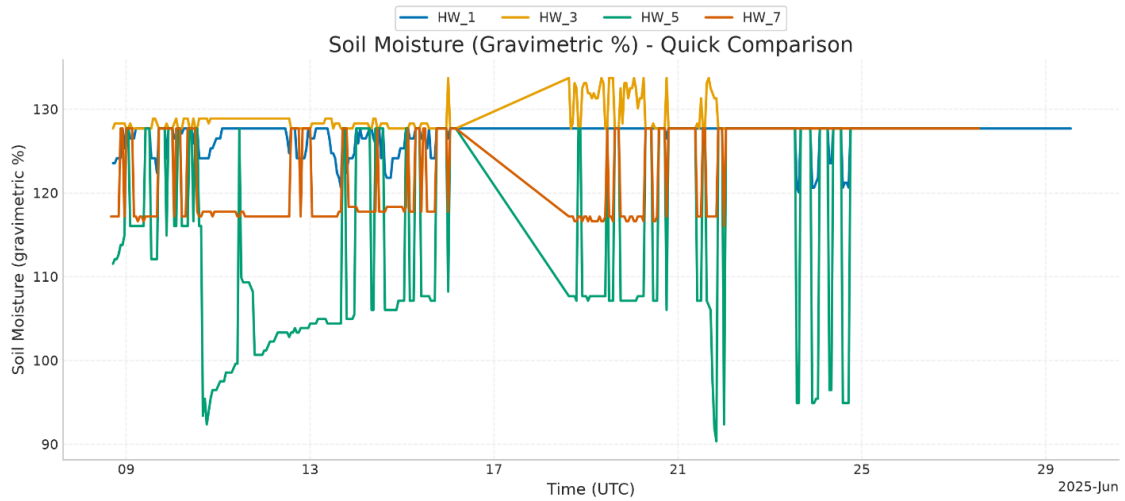


Figure 4.4: Soil moisture measurements during the preliminary field deployment.

deployment successfully demonstrated that the sensing hardware, analog measurement circuitry, and cloud logging system operated correctly under outdoor conditions.

4.3.3. Summary of Preliminary Testing

The preliminary field deployment successfully fulfilled its intended purpose as an engineering validation phase. Although the collected environmental data were not suitable for scientific analysis due to communication interruptions and ongoing hardware development, the testing identified several important issues that were subsequently corrected. Most notably, replacing the incorrectly matched antennas with antennas specifically designed for the 433 MHz LoRa frequency significantly improved communication reliability and packet delivery in later deployments. Additional improvements included refinements to the firmware, sensor calibration procedures, enclosure design, and overall system stability.

The knowledge gained from this preliminary deployment formed the basis for the final hardware revision used in the subsequent long-term field deployment, where substantially improved communication reliability and data quality were achieved.

4.4. LoRa Range Testing Results (Vellir – Hafnarfjörður)

4.4.1. Test Configurations

The following configuration parameters were used during the LoRa field range testing:

```
#define FREQUENCY       433E6   // 433 MHz
#define TX_POWER        17      // dBm
#define BANDWIDTH       125E3   // 125 kHz
#define SYNC_WORD       0x12
#define SPREAD_FACTOR   10
#define CODING_RATE     5
#define PREAMBLE        8
```

During the initial field test, communication performance was limited due to the use of antennas designed for Wi-Fi rather than LoRa. These antennas were not tuned for the 433 MHz operating frequency, resulting in significantly reduced range. After replacing them with antennas designed specifically for LoRa frequencies, a substantial improvement in communication performance was observed.

Field testing was conducted in a semi-urban environment around Vellir in Hafnarfjörður. Using the configuration described above, reliable communication was achieved within an approximate radius of 3 km. Signal quality improved when the transmitting node was positioned at elevated locations, highlighting the importance of antenna placement.

In areas with dense infrastructure, such as residential zones with multiple buildings obstructing the line of sight, signal degradation and occasional packet loss were observed. This confirms that LoRa performance is strongly influenced by environmental obstructions and benefits from relatively clear propagation paths.

The estimated signal coverage area derived from the field test is shown in the estimated coverage map. Regions marked in red represent areas where no signal was detected, primarily due to terrain obstructions such as hills and dense built environments.

Signal strength measurements were recorded at multiple locations within the testing area. The average RSSI values are summarized in the RSSI table.

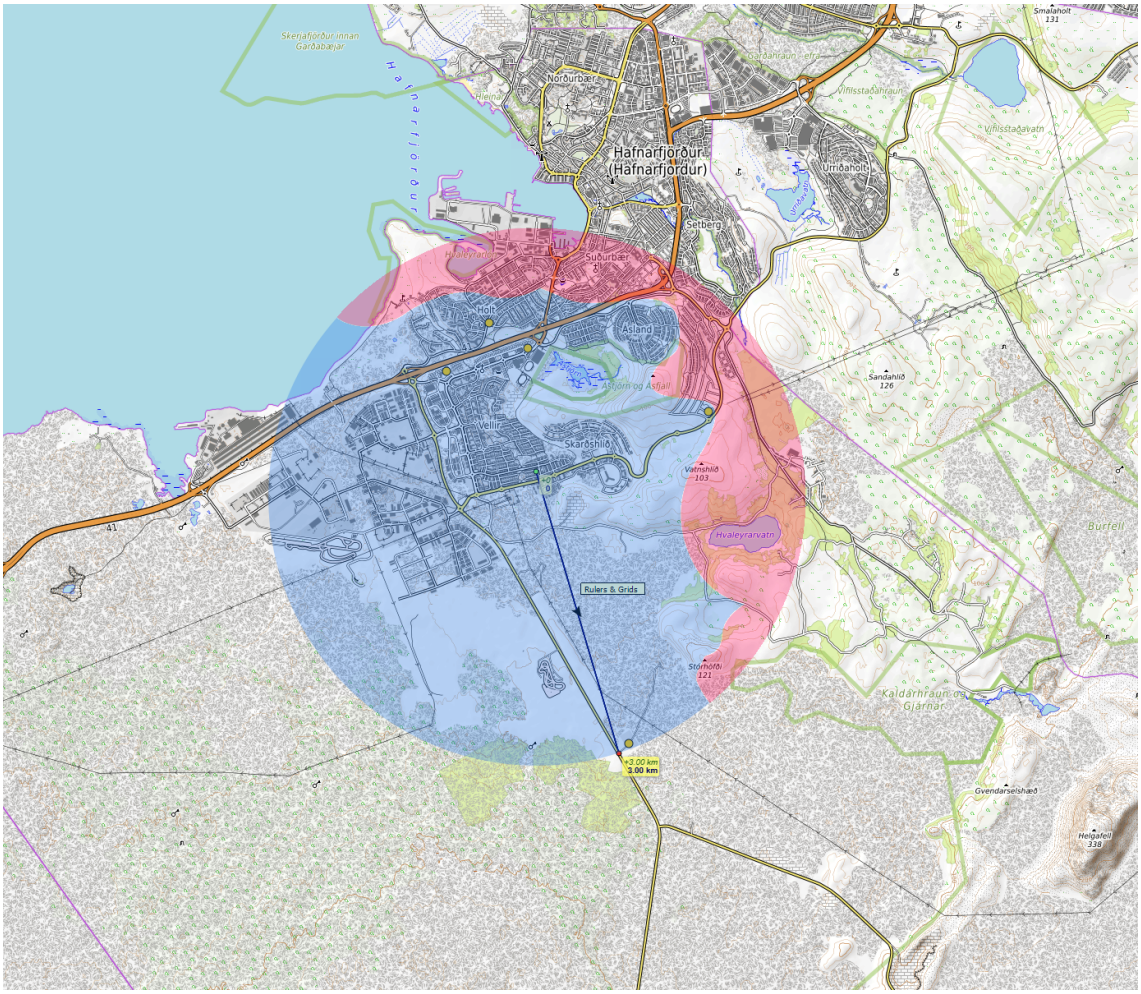


Figure 4.5: Estimated LoRa signal coverage map for the Vellir–Hafnarfjörður field test. Blue regions indicate reliable communication coverage at approximately 3 km, while red regions indicate weak or absent signal strength due to terrain and structural obstructions.

Table 4.2: Average RSSI values recorded at each test location during the LoRa range evaluation.

Location	**Average RSSI**
Ásvallalaug	-113 dBm
Holt	-103 dBm
Fish Drying Area	-112 dBm
Vatnslíð	-103 dBm
Netto	-112 dBm

4.4.2. Discussion of Results

The results demonstrate that the system achieves an approximate communication radius of 3 km, even with non-ideal base station placement. The recorded RSSI values, ranging from -103 dBm to -113 dBm, indicate that communication remains stable near the sensitivity limits of the receiver.

Although mesh networking was not implemented during this test, the results highlight its potential advantages. In a mesh topology, intermediate nodes can relay data, effectively extending the communication range beyond the limits observed in a single-hop configuration.

For Icelandic turf house applications, where structures are typically located within short distances of each other, a 3 km communication range is more than sufficient for reliable node-to-node communication. This supports the feasibility of deploying multiple sensor nodes across sites while maintaining connectivity through a mesh network.

Furthermore, the system allows flexibility in base station placement. The base station can be located in areas with available internet connectivity, while remote nodes transmit data through intermediate nodes until reaching the gateway.

Despite additional attenuation from soil burial and seasonal snow cover, the observed signal strength suggests that communication can still be maintained under such conditions. This is critical for the intended application, as the sensor nodes are deployed beneath turf roofs to monitor soil conditions.

Overall, the results confirm that LoRa technology is well suited for long-range, low-power environmental monitoring in Icelandic turf houses, providing reliable communication even under challenging environmental conditions.

4.5. Intermediate Field Deployment

Following the preliminary field testing, an intermediate field deployment was conducted to evaluate the performance of the revised hardware after replacing the original antennas with antennas specifically designed for the **433 MHz LoRa operating frequency**. The deployment was again carried out at several turf roof locations around the **University of Iceland (Háskóli Íslands)** campus, allowing rapid access to the sensor nodes for hardware modifications and firmware updates while operating under real outdoor conditions.

The primary objective of this deployment was to verify the improvements in LoRa communication reliability, evaluate the stability of the environmental sensors, and identify any remaining hardware or software issues before proceeding to the final long-term deployment. Four sensor nodes (HW_1, HW_3, HW_5, and HW_7) continuously monitored air temperature, relative humidity, soil temperature, and soil moisture, while measurements from the nearby weather station were used as a reference.

4.5.1. Air Temperature and Relative Humidity

Figure 4.6 compares the measured air temperature from the four prototype sensor nodes with the nearby weather station, while Figure 4.29 presents the corresponding relative humidity measurements.

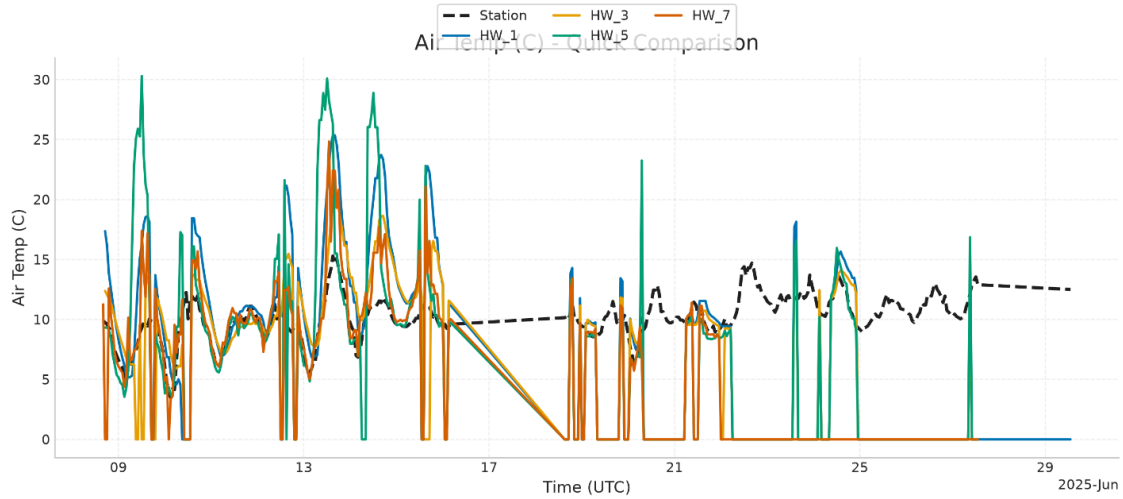


Figure 4.6: Air temperature measurements during the intermediate field deployment.

Compared with the preliminary deployment described in Section 4.3, replacing the antennas with properly matched **433 MHz LoRa antennas** resulted in a noticeable improvement in communication reliability. Longer periods of uninterrupted environmental data were successfully transmitted to the base station, reducing the amount of missing data caused by packet loss.

The measured air temperatures generally followed the same trends as the nearby weather station, demonstrating that the revised communication hardware allowed more consistent data collection. Relative humidity measurements also became more stable; however, occasional readings of **0%** and **100%** were still observed. These values were attributed primarily to ongoing firmware development and sensor initialization issues rather than communication failures. Since the system remained

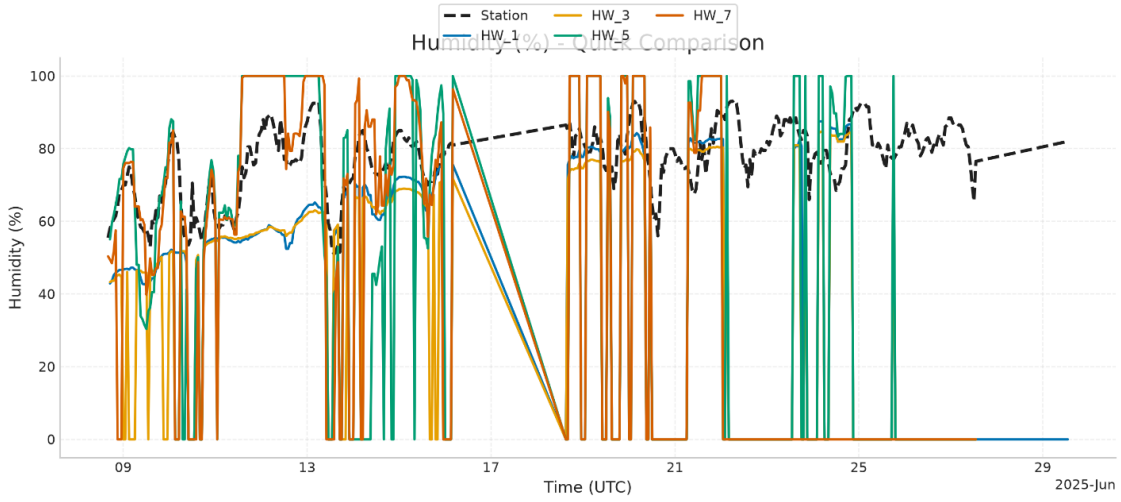


Figure 4.7: Relative humidity measurements during the intermediate field deployment.

under active development, firmware updates and hardware adjustments continued throughout the testing period.

4.5.2. Soil Temperature and Soil Moisture

Figure 4.8 presents the measured soil temperature from the deployed sensor nodes, while Figure 4.9 illustrates the corresponding soil moisture measurements.

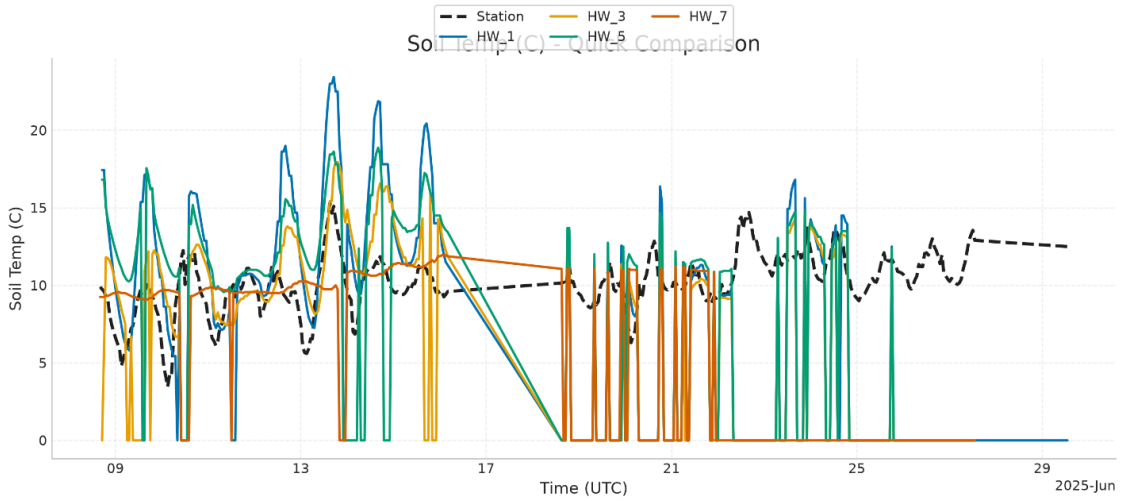


Figure 4.8: Soil temperature measurements during the intermediate field deployment.

Soil temperature measurements became more continuous than those obtained during the preliminary deployment, reflecting the improved communication performance

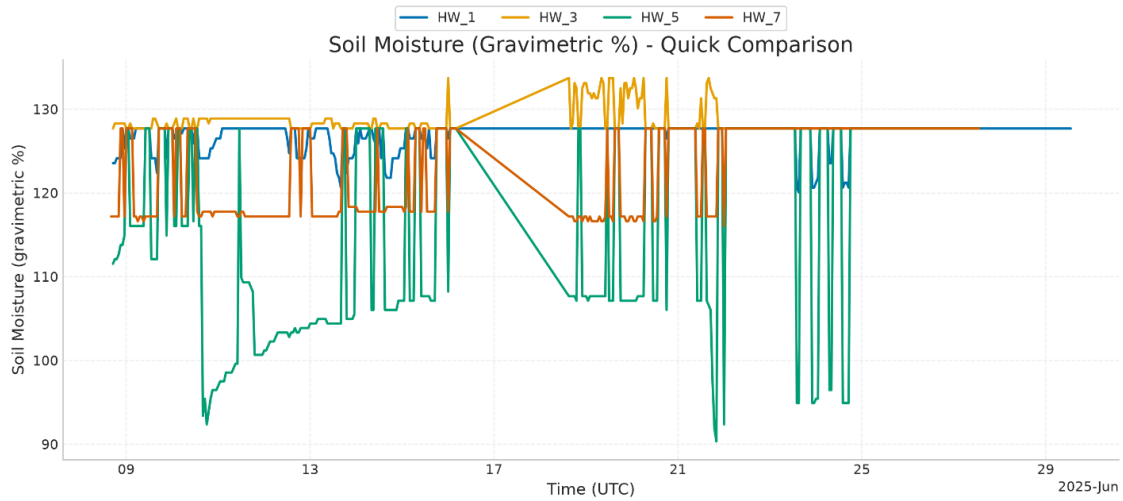


Figure 4.9: Soil moisture measurements during the intermediate field deployment.

achieved after replacing the antennas. The measured soil temperatures remained more stable than ambient air temperatures, demonstrating the thermal buffering effect of the surrounding soil.

The soil moisture measurements continued to exhibit considerable variability because calibration procedures were still being refined. Several nodes produced unrealistic values outside the expected measurement range, indicating that further calibration and firmware improvements were required before long-term deployment. Although these measurements were not yet suitable for quantitative environmental analysis, they confirmed that the sensing hardware, analog measurement circuitry, and data logging system were operating correctly under outdoor conditions.

4.5.3. Summary

The intermediate field deployment confirmed that replacing the original antennas with antennas specifically designed for the **433 MHz LoRa frequency** substantially improved wireless communication reliability. Compared with the preliminary deployment, significantly longer periods of continuous environmental data were successfully collected, demonstrating that the antenna mismatch had been a major contributor to the communication problems previously observed.

Although several issues related to firmware stability and sensor calibration remained, the deployment successfully validated the revised communication hardware and identified the final improvements required before the long-term deployment. The experience gained during this stage led to further refinements in the firmware,

sensor calibration procedures, and enclosure design, resulting in the final hardware configuration used for the subsequent deployment at the Keldur turf house site.

4.6. Keldur Deployment

To evaluate the performance of the proposed LoRa mesh wireless sensor network under actual operating conditions, a field deployment was conducted at the Keldur turf house site beginning on **17 October 2025**. Four sensor nodes (IDs 1, 3, 5, and 7) were installed at different locations and depths beneath the turf roof. The installation locations and sensor depths are shown in **the deployment map**. Sensor Nodes 1 and 3 were designed with the air temperature and humidity sensors enclosed inside the waterproof housing, while Sensor Nodes 5 and 7 had these sensors mounted externally above the enclosure. This design variation was intentionally implemented to compare the measurement accuracy of both configurations under real environmental conditions.

4.6.1. Air Temperature Performance

Figure 4.11 compares the measured air temperature from all deployed sensor nodes with data obtained from the nearby weather station.

The results show that Sensor Nodes 5 and 7 closely followed the weather station throughout the deployment period. Daily temperature fluctuations, cold periods, and warming trends were clearly captured by both sensors with only minor deviations. This indicates that placing the air temperature sensor outside the enclosure provides measurements that better represent the surrounding atmospheric conditions.

In contrast, Sensor Nodes 1 and 3, whose air sensors were installed inside the waterproof enclosure, produced smoother temperature curves with noticeably reduced daily fluctuations. Internal heating from electronic components together with the thermal insulation provided by the enclosure resulted in a damped response compared to the weather station. Although these measurements remained useful for monitoring the internal sensor environment, they were less representative of the true ambient air temperature.

The average absolute temperature difference between the deployed sensors and the weather station ranged from approximately **3.06 °C to 3.27 °C**, with Sensor Node 7 producing the lowest average deviation among all deployed units. Overall, the

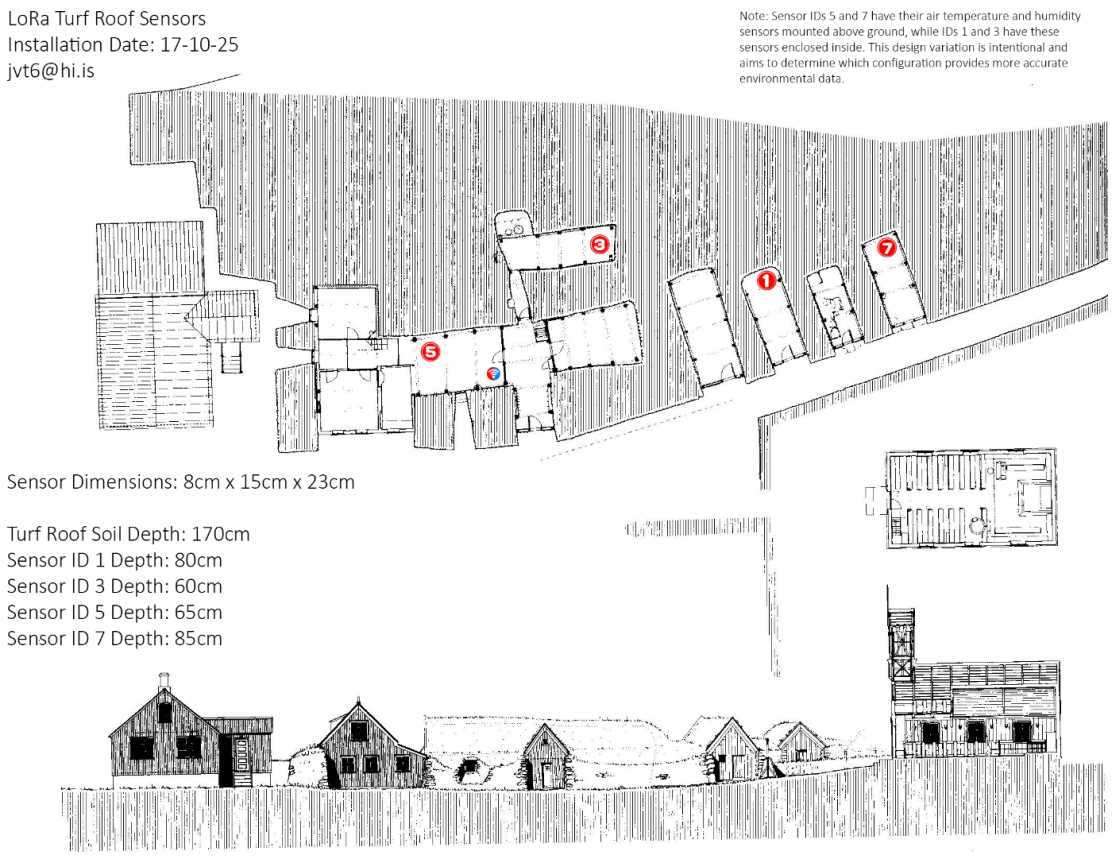


Figure 4.10: Deployment map showing the locations and installation depths of the four sensor nodes at the Keldur turf roof site.

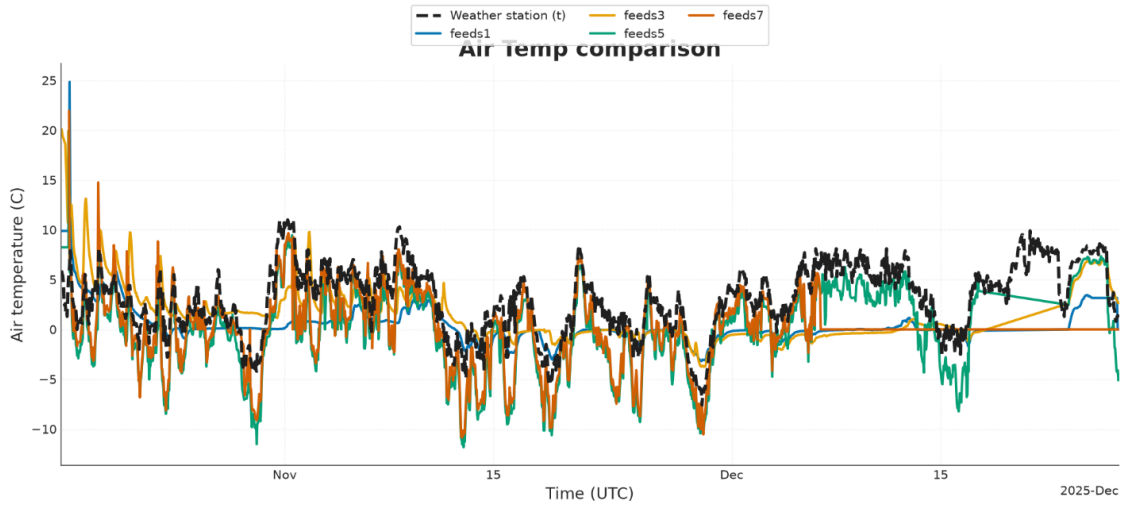


Figure 4.11: Comparison of air temperature measurements from the deployed sensor nodes and the nearby weather station.

results indicate that external placement of the air temperature sensor is preferable when accurate ambient air measurements are required.

4.6.2. Relative Humidity Performance

Figure 4.12 presents the relative humidity measurements obtained from all deployed nodes together with the weather station reference.

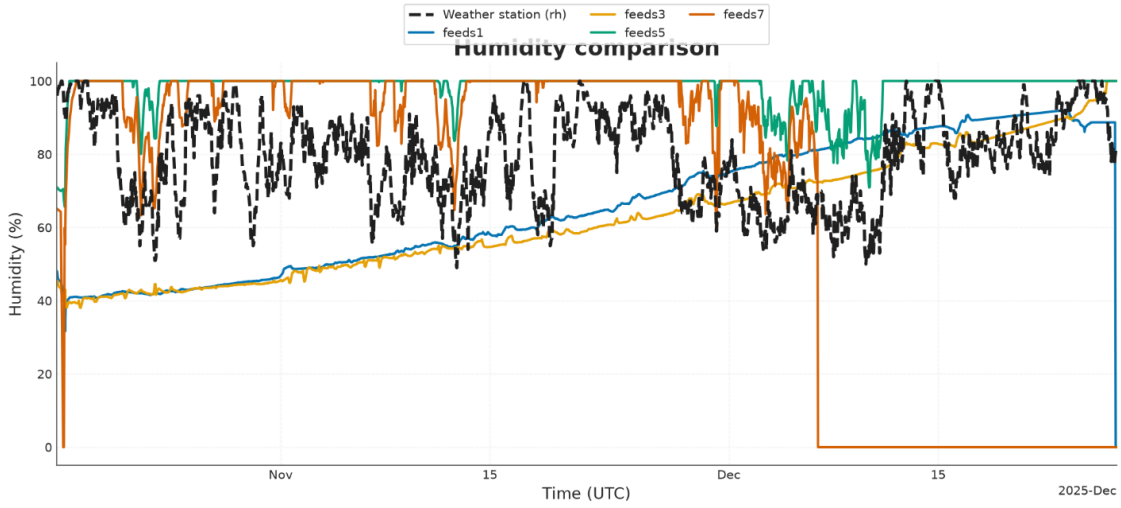


Figure 4.12: Comparison of relative humidity measurements from the deployed sensor nodes and the nearby weather station.

Sensor Nodes 5 and 7, with externally mounted humidity sensors, closely tracked the overall humidity changes observed by the weather station. High humidity events approaching saturation and periods of decreasing humidity were successfully detected throughout the deployment.

Conversely, Sensor Nodes 1 and 3, which housed the humidity sensors inside the enclosure, displayed a gradual increase in humidity over time. Instead of rapidly responding to environmental changes, the enclosed sensors accumulated moisture inside the housing, resulting in steadily increasing humidity measurements. This confirms that the waterproof enclosure significantly influences humidity measurements by restricting air exchange with the surrounding environment.

Sensor Node 7 experienced a hardware failure during December, after which humidity readings became fixed at 0%, clearly indicating sensor malfunction rather than actual environmental conditions. Despite these issues, the results demonstrate that externally mounted humidity sensors provide measurements that more accurately represent ambient conditions.

4.6.3. Soil Temperature Performance

Figure 4.13 illustrates the soil temperature measurements collected by the four buried sensor nodes together with the available temperature data from the nearby weather station.

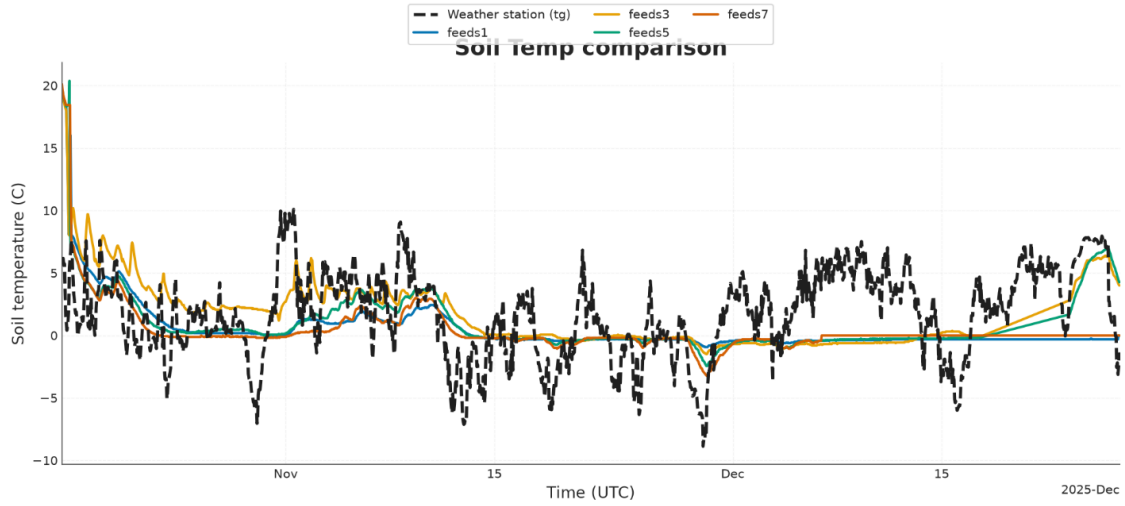


Figure 4.13: Comparison of soil temperature measurements from the deployed sensor nodes and the nearby weather station.

Unlike air temperature, soil temperature measurements from all four deployed nodes exhibited similar behavior regardless of installation configuration. The measured soil temperatures were significantly more stable than the weather station temperature, remaining close to the freezing point for much of the observation period. Rapid daily fluctuations observed by the weather station were heavily damped by the thermal mass of the turf roof and surrounding soil.

This behavior is expected since soil naturally buffers rapid temperature changes. Even when air temperatures varied between approximately **−8 °C and +10 °C**, soil temperatures remained close to **0 °C** with only gradual seasonal variation.

It should be noted that the nearby weather station did **not** provide underground soil temperature measurements at the same depth as the deployed sensors. As a result, the comparison was performed using the weather station's **ground-level (surface) temperature** data, which served as the closest available reference. Since the developed sensor nodes measured soil temperature at depths ranging from approximately **60 mm to 85 mm** below the turf surface, some difference between the two datasets was expected. Soil temperatures below the surface are naturally more thermally stable and less affected by rapid atmospheric temperature changes than temperatures measured at or near the ground surface.

Consequently, the average absolute difference between the deployed sensor measurements and the weather station data ranged from approximately **2.77 °C to 3.11 °C**. This difference should therefore not be interpreted solely as measurement error but rather as a consequence of comparing temperatures measured at different depths and under different environmental conditions.

Overall, the soil temperature measurements demonstrated consistent long-term behavior and confirmed that the sensors operated reliably under buried conditions.

4.6.4. Individual Sensor Performance

Figures 4.24 to 4.27 present the complete time-series measurements recorded by each deployed sensor node, including air temperature, relative humidity, soil temperature, soil moisture, and battery voltage throughout the deployment period.

Sensor Node 1 Sensor Node 1 maintained stable communication throughout most of the deployment while continuously monitoring environmental conditions. Soil moisture measurements remained relatively low with gradual increases following wetter periods before decreasing as drainage occurred. Battery voltage gradually decreased from approximately **4.1 V** to **3.7 V**, demonstrating efficient power consumption. Near the end of the deployment, the node experienced a sudden loss of power, resulting in the termination of data collection.

Sensor Node 3 Sensor Node 3 operated reliably throughout the deployment and exhibited the most complete dataset among all deployed nodes. Soil moisture increased during wetter conditions before gradually decreasing as the roof dried. Battery voltage showed a smooth and consistent discharge profile, indicating stable operation and validating the effectiveness of the implemented low-power algorithm.

Sensor Node 5 Sensor Node 5 produced air temperature and humidity measurements that closely matched the nearby weather station due to its externally mounted atmospheric sensors. Soil moisture remained consistently higher than the other deployed nodes, suggesting that its installation location retained more water within the turf roof. Battery voltage remained nearly constant throughout the deployment, demonstrating excellent energy efficiency despite continuous operation.

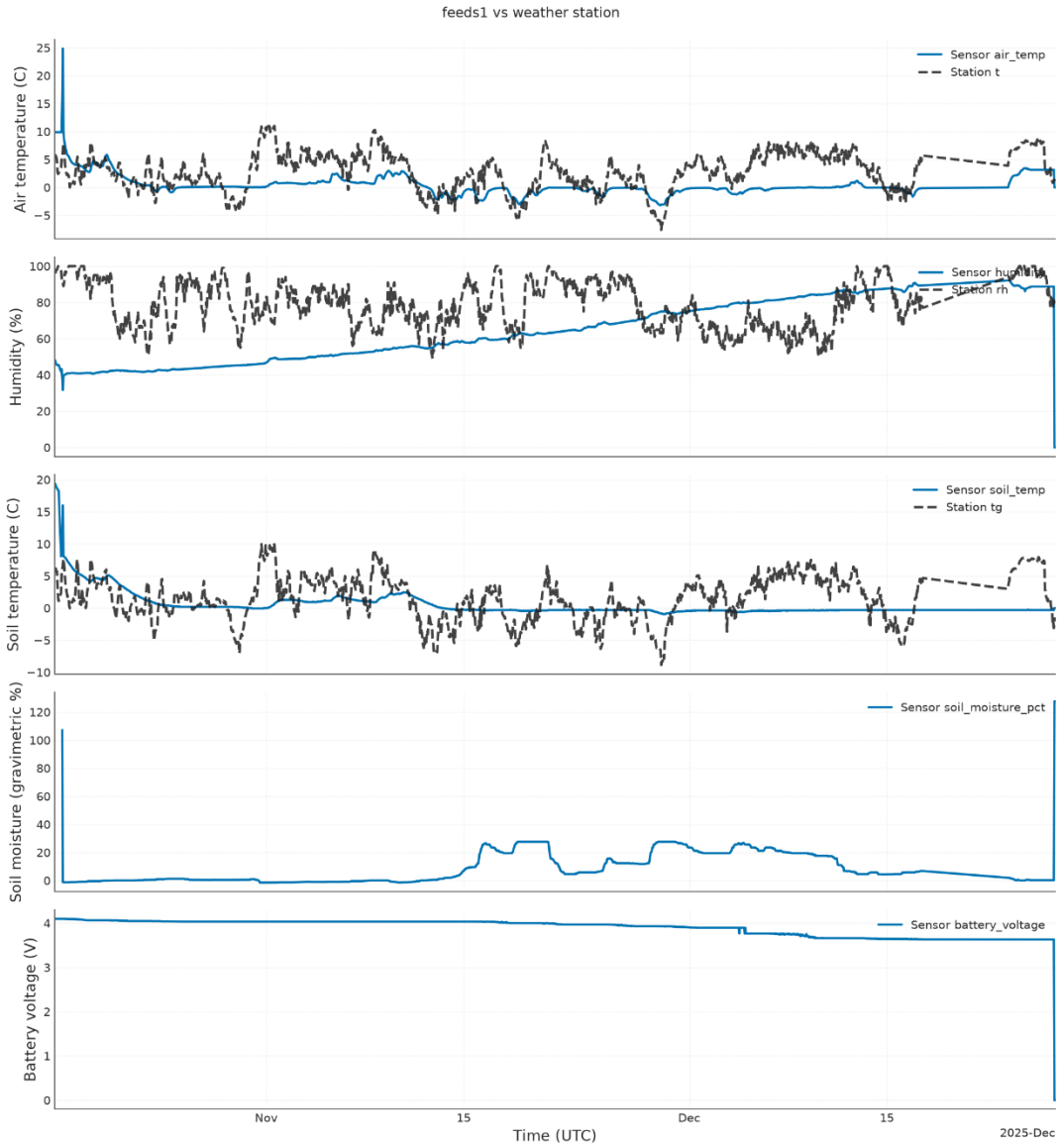


Figure 4.14: Environmental measurements recorded by Sensor Node 1 throughout the deployment period.

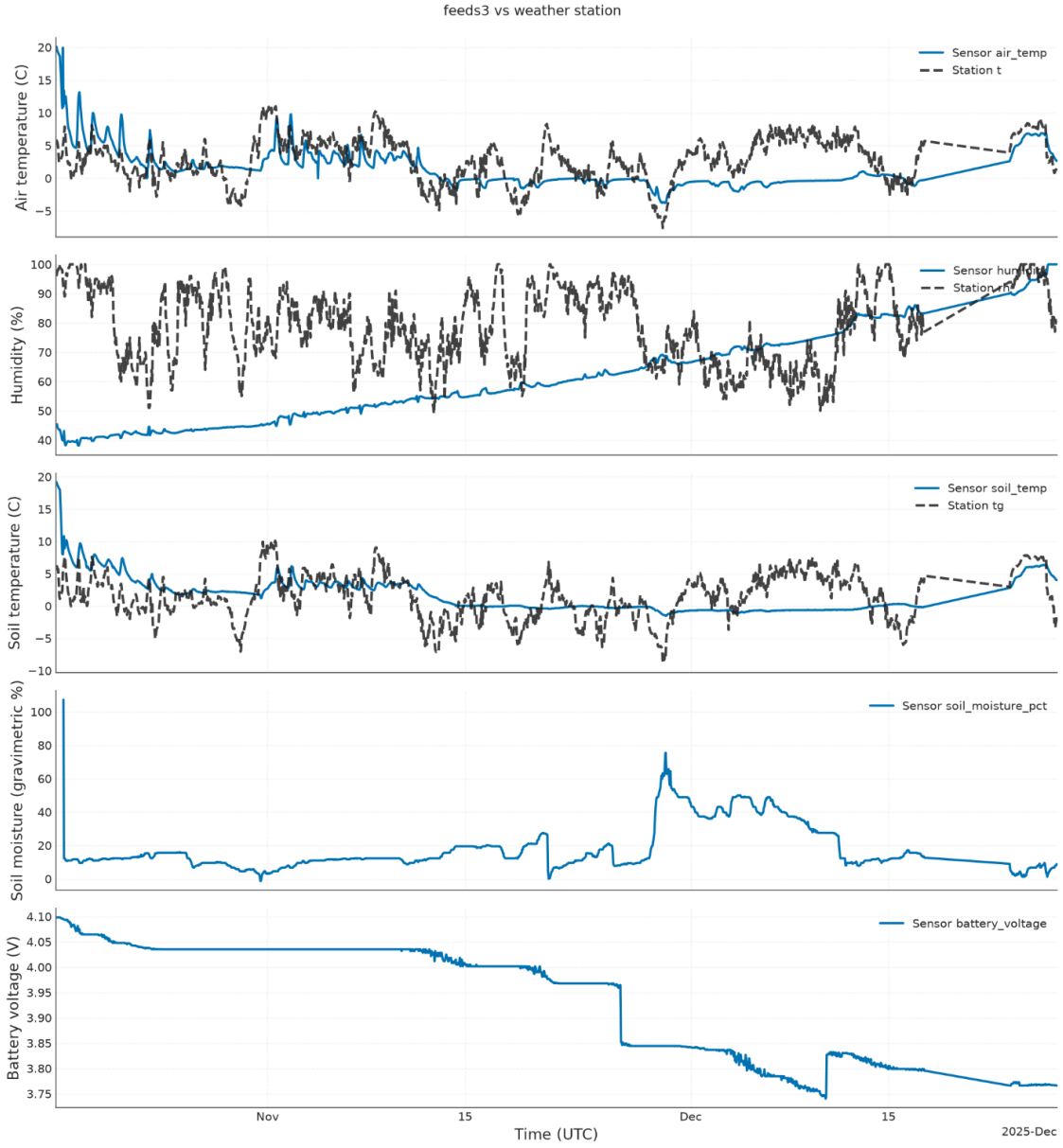


Figure 4.15: Environmental measurements recorded by Sensor Node 3 throughout the deployment period.

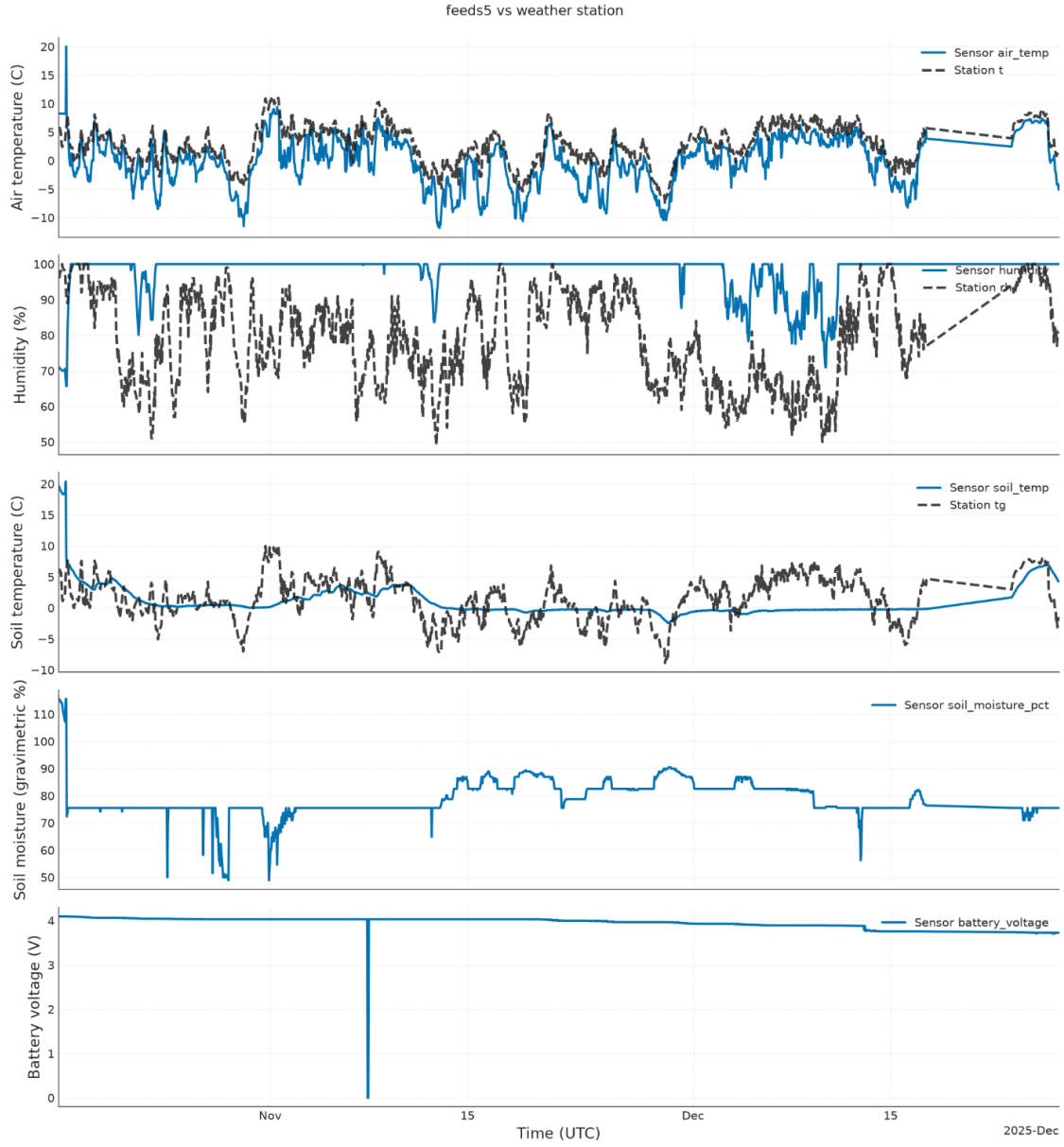


Figure 4.16: Environmental measurements recorded by Sensor Node 5 throughout the deployment period.

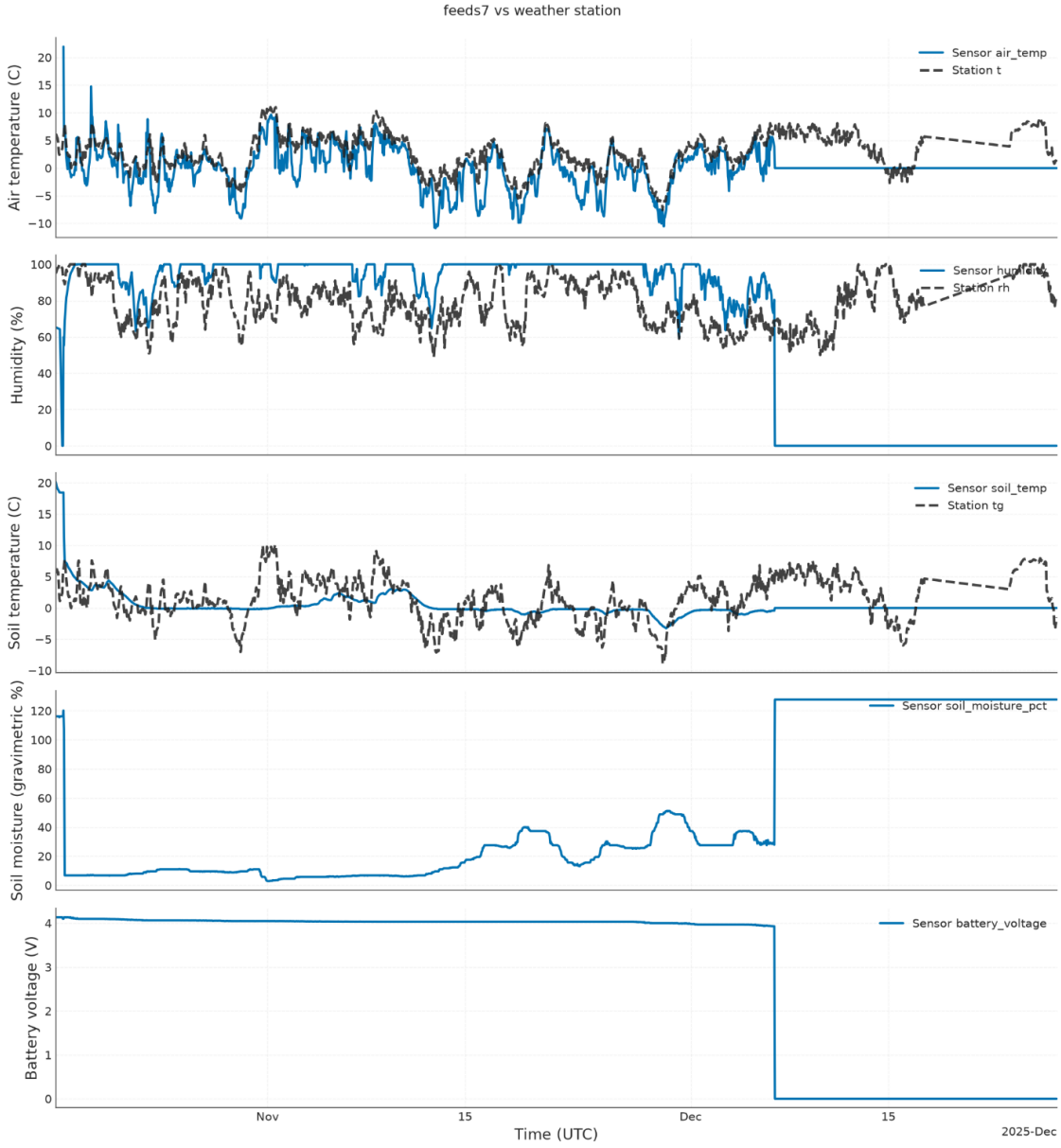


Figure 4.17: Environmental measurements recorded by Sensor Node 7 throughout the deployment period.

Sensor Node 7 Sensor Node 7 initially produced measurements comparable to Sensor Node 5, with externally mounted atmospheric sensors closely following weather station observations. However, a hardware or power failure occurred during December, after which battery voltage dropped to zero. This resulted in invalid humidity measurements fixed at 0% and soil moisture readings exceeding 100%, indicating sensor failure rather than actual environmental conditions. Despite this failure, the node successfully demonstrated stable operation during the majority of the deployment period.

4.6.5. Overall System Performance

The field deployment demonstrated that the proposed LoRa mesh wireless sensor network successfully monitored environmental conditions on an Icelandic turf roof under real operating conditions. Throughout the deployment, the system continuously measured air temperature, relative humidity, soil temperature, soil moisture, and battery voltage while transmitting data to the cloud for remote monitoring.

Comparison with the nearby weather station showed that externally mounted atmospheric sensors provided significantly better agreement with ambient environmental conditions than sensors installed inside the waterproof enclosure. Soil temperature measurements remained stable despite large fluctuations in air temperature, demonstrating the thermal buffering characteristics of the turf roof. Battery monitoring further confirmed the effectiveness of the implemented low-power design, with most sensor nodes maintaining stable battery voltage throughout more than two months of continuous operation.

Although two sensor nodes experienced hardware failures near the end of the deployment, the remaining nodes continued operating successfully, demonstrating the robustness of the proposed monitoring system under real outdoor conditions. Overall, the field deployment validates the feasibility of using a low-power LoRa mesh wireless sensor network for long-term monitoring of soil conditions on Icelandic turf roofs.

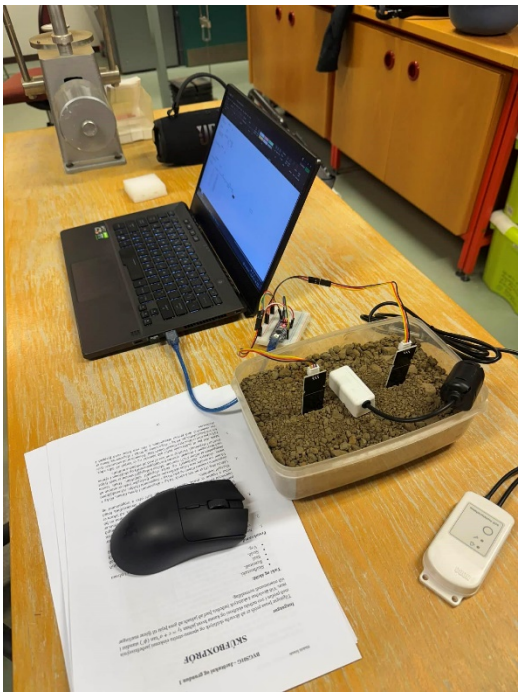
4.7. Soil Moisture Calibration Results

(Gravimetric Method vs HOBO MX2307 vs Custom Sensor)

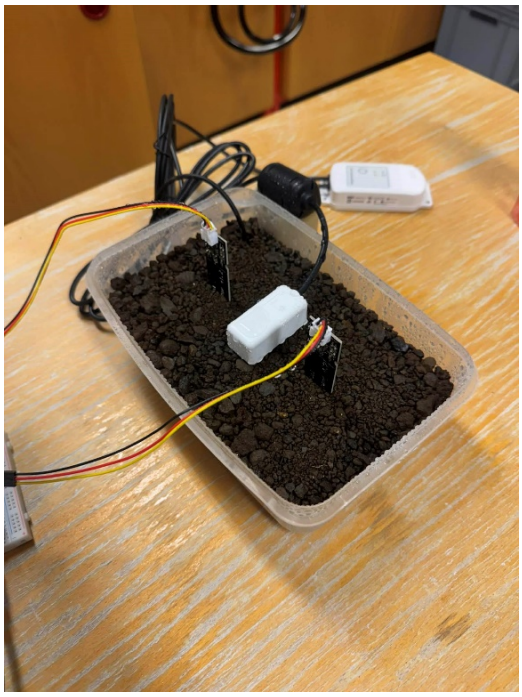
4.7.1. Gravimetric Reference Data

Soil moisture calibration was performed using controlled samples at 0%, 10%, 20%, 30%, and 40% moisture content. The gravimetric method was used as the reference by adding known volumes of water to a measured mass of dry soil.

Calibration was limited to 40% moisture, as the soil approached saturation beyond this point. Additional water produced minimal dielectric variation, reducing the sensitivity of capacitive measurements.



(a) Calibration setup with the soil sample, custom sensors, and reference instrument.



(b) Custom sensors installed in the prepared soil sample.

Figure 4.18: Laboratory setup used for the soil moisture sensor calibration tests.

Two calibration tests were conducted.

Test 1 (2069.5 g soil sample):

Table 4.3: Calibration Test 1 results for the 2069.5 g soil sample.

Soil Moisture (%)	**Water Added (mL)**	**VWC (m^3/m^3)**	**Raw Sensor 1**	**Raw Sensor 2**
0	0	0	483.52	486.26
10	206.95	0.15	314.17	324.55

Table 4.3: Calibration Test 1 results for the 2069.5 g soil sample.

Soil Moisture (%)	Water Added (mL)	VWC (m^3/m^3)	Raw Sensor 1	Raw Sensor 2
20	413.9	0.27	232.17	239.29
30	620.85	0.33	223.26	216.67
40	827.8	0.39	193.20	201.16

The large sample size resulted in long saturation times and difficult handling.

Test 2 (400 g soil sample):

Table 4.4: Calibration Test 2 results for the 400 g soil sample.

Soil Moisture (%)	Water Added (mL)	VWC (m^3/m^3)	Raw Sensor 1
0	0	0.02	482.55
10	40	0.21	260.36
20	80	0.29	265.29
30	120	0.34	220.68
40	160	0.39	202.52

The second test improved practicality while maintaining consistent trends. In both tests, ADC values decreased as moisture increased, confirming expected capacitive sensor behavior.

4.7.2. HOBO Sensor and Dielectric Reference

A HOBO MX2307 (TEROS-10 probe) was used as a reference. It measures dielectric permittivity and converts it to volumetric water content (VWC) using:

$$\theta = 4.3 \times 10^{-6}\varepsilon^3 - 5.5 \times 10^{-4}\varepsilon^2 + 2.92 \times 10^{-2}\varepsilon - 5.3 \times 10^{-2} \tag{4.12}$$

Measured VWC values were back-calculated to obtain dielectric permittivity:

Table 4.5: HOBO reference measurements and calculated dielectric permittivity.

Moisture (%)	**HOBO VWC (m^3/m^3)**	**Dielectric Permittivity (ε)**
0	0.01	2.251
10	0.18	9.578
20	0.28	15.273
30	0.335	19.175
40	0.39	23.957

Dielectric permittivity increased with moisture due to the high dielectric constant of water.

4.7.3. Polynomial Calibration Model

Due to the nonlinear relationship between ADC output and soil moisture, a second-order polynomial regression was applied:

$$\text{Soil Moisture (\%)} = 0.0006877(\text{ADC})^2 - 0.5961(\text{ADC}) + 127.70 \tag{4.13}$$

The model was derived from averaged calibration data:

Table 4.6: Averaged calibration data used to derive the polynomial model.

Moisture (%)	**Avg VWC**	**Raw Avg ADC**
0	0.01	484.11
10	0.18	299.69
20	0.28	245.58
30	0.335	220.20
40	0.39	198.96

4.7.4. Final Calibration Results and Discussion

Table 4.7: Final comparison of gravimetric, HOBO, dielectric, and calibrated moisture values.

Gravimetric (%)	ADC	HOBO VWC	Dielectric (ε)	Calibrated (%)
0	484.89	0.01	2.251	~0
10	319.36	0.18	9.578	~11
20	235.73	0.28	15.273	~20
30	219.96	0.335	19.175	~28
40	197.18	0.39	23.957	~38

The custom sensor reliably measured soil moisture within the 0–40% range. The response was nonlinear, with higher sensitivity at low moisture levels and reduced sensitivity near saturation, consistent with capacitive sensing principles.

Comparison with the HOBO MX2307 showed good agreement, with an approximate root mean square error of **~3% soil moisture**.

While the HOBO system provides higher absolute accuracy, the custom sensor is sufficiently accurate for long-term monitoring of moisture trends in Icelandic turf roofs. Detecting wetting and drying cycles is more critical than achieving laboratory-grade precision.

Overall, the calibrated sensor provides reliable performance for integration into the low-power LoRa-based monitoring system.

4.8. Cloud Upload Performance (ThingSpeak)

The cloud upload functionality was evaluated using the ThingSpeak IoT platform. Sensor data from the base station—including soil moisture, temperature, humidity, battery voltage, and timestamps—was transmitted via Wi-Fi using HTTP POST requests. During testing, the system operated reliably with no failed transmissions or corrupted data. All uploads were successfully received by the server, consistently returning an HTTP response code of *200 OK*. The average upload time was approximately 1–2 seconds depending on network conditions, with an observed failure rate of 0%.

These results indicate that the cloud communication system is stable and suitable for long-term environmental monitoring. The Wi-Fi connection remained consistent, and no missed uploads were observed, demonstrating reliable integration between LoRa

data collection and cloud storage. Given that the application prioritizes periodic monitoring rather than high data throughput, the achieved performance is more than sufficient. Furthermore, integration with ThingSpeak enables convenient remote access, visualization, and analysis of environmental data from Icelandic turf roofs.

5. Conclusion and Recommendations

The completed implementation demonstrates the feasibility of a low-cost, low-power LoRa wireless sensor mesh network for monitoring environmental conditions beneath Icelandic turf roofs. The system integrates distributed sensing, local data storage, LoRa communication, a base station, and cloud storage into a single monitoring architecture.

The laboratory power measurements indicate an average current consumption of approximately 3.08 mA under the documented worst-case operating cycle. With a nominal 10,500 mAh battery capacity, the idealized calculated lifetime is approximately 142 days. The thesis results appropriately note that low temperatures and other environmental factors can reduce the practical battery capacity and therefore shorten the actual lifetime.

Wireless testing showed that antenna selection had a major effect on system performance. Early field testing suffered from packet loss because the deployed antennas were not properly matched to the 433 MHz LoRa frequency. Replacing them with suitable LoRa antennas substantially improved communication reliability. Range testing around Vellir in Hafnarfjörður subsequently demonstrated reliable communication within an approximate 3 km radius, with measured RSSI values between approximately -103 dBm and -113 dBm. The results support the use of mesh relaying for applications where individual nodes cannot directly reach the base station.

The Keldur deployment demonstrated that the system can collect and transmit environmental measurements under actual turf-roof conditions. The deployment continuously monitored air temperature, relative humidity, soil temperature, soil moisture, and battery voltage. Two nodes experienced hardware failures near the end of the documented period, but the remaining nodes continued to operate, demonstrating useful system-level resilience while also identifying the need for further hardware reliability improvements.

Soil moisture calibration established a nonlinear relationship between the low-cost capacitive sensor output and reference moisture measurements over the 0–40% range. The resulting second-order calibration model achieved an approximate RMSE of 3% soil moisture relative to the HOBO reference. This level of performance is considered

suitable for monitoring wetting and drying trends, although the calibrated range should be expanded if measurements near saturation are required.

Finally, the ThingSpeak integration successfully uploaded sensor measurements using HTTP POST requests. The documented tests achieved HTTP 200 responses, an average upload time of approximately 1–2 seconds, and an observed failure rate of 0%. This confirms that the cloud component provides a practical method for remote access, visualization, and storage of the collected measurements.

5.1. Recommendations

Future work should focus on extending the duration of field deployments, improving the reliability of individual sensor nodes, and expanding calibration and validation. In particular, the following improvements are recommended:

1. improve battery and power-management design to increase practical lifetime under Icelandic winter temperatures;

2. strengthen enclosure and cable protection for long-term exposure to moisture, freezing, thawing, and snow;

3. investigate the causes of the observed node failures and add monitoring mechanisms that can identify failed or low-voltage nodes remotely;

4. extend soil-moisture calibration beyond the current 0–40% range where physically meaningful and validate the calibration using additional soil samples and reference measurements;

5. conduct dedicated multi-hop mesh-range tests to quantify the benefit of relay nodes rather than relying only on single-hop range measurements;

6. refine the external placement of the air temperature and humidity sensors, since the results indicate that external placement better represents ambient conditions; and

7. continue long-term deployment at additional turf-roof sites to evaluate repeatability across different roof structures, soil conditions, and seasonal conditions.

References

[1] H. Stefánsson. *H. Stefánsson*. 2013.

[2] UNESCO World Heritage Centre. *Turf House Tradition,* 2011. URL: https://whc.unesco.org/en/tentativelists/5589/.

[3] A. Robock. *HYDROLOGY | Soil Moisture,* 2003. DOI: 10.1016/B0-12-227090-8/00169-X.

[4] L. Zhang et al. *In situ observation-constrained global surface soil moisture using random forest model,* 2021. DOI: 10.3390/rs13234893.

[5] G. Mander and M. Arora. *design of capacitive sensor for monitoring moisture content of soil and analysis of analog voltage with variability in moisture*. 2014. DOI: 10.1109/raecs.2014.6799646.

[6] University of Massachusetts. *Measuring Soil Moisture,* 2002. URL: https://ag.umass.edu/fact-sheets/measuring-soil-moisture.

[7] M. Hartings et al. *Humidity responsive photonic sensor based on a carboxymethyl cellulose mechanical actuator*. 2017. DOI: 10.26434/chemrxiv.5492728.v1.

[8] *What is humidity?,* 2023. URL: https://scijinks.gov/what-is-humidity/.

[9] M. Hartings et al. *Humidity responsive photonic sensor based on a carboxymethyl cellulose mechanical actuator*. 2017. DOI: 10.26434/chemrxiv.5492728.v1.

[10] C. W. Callahan, A. M. Elansari, and D. L. Fenton. *Psychrometrics,* 2019. DOI: 10.1016/B978-0-12-813276-0.00008-0.

[11] R. Bathurst, D. Zori, and J. Byock. *Diatoms as bioindicators of site use: locating turf structures from the viking age,* 2010. DOI: 10.1016/j.jas.2010.07.002.

[12] S. Nannini. *Icelandic farmhouses,* 2023. DOI: 10.36253/979-12-215-0084-4.

[13] B. Lin et al. *Unlocking the water retention behaviour of turf construction materials,* 2023. DOI: 10.1051/e3sconf/202338217008.

[14] T. Wunderlich et al. *On the ability of geophysical methods to image medieval turf buildings in Iceland,* 2015. DOI: 10.1002/arp.1506.

[15] S. Sigurðardóttir. *S. Sigurðardóttir*. 2008.

[16] H. Lárusson. *The Icelandic Farmstead,* 2016.

[17] J. Van Hoof and F. Van Dijken. *The Historical Turf Farms of Iceland: Architecture, Building Technology and the Indoor Environment,* 2008.

[18] S. B. Hafsteinsson. *Museum Politics and Turf-House Heritage,* 2010.

[19] K. Milek and H. Roberts. *Integrated geoarchaeological methods for the determination of site activity areas: a study of a viking age house in Reykjavik, Iceland,* 2013. DOI: 10.1016/j.jas.2012.10.031.

[20] K. Milek. *Floor formation processes and the interpretation of site activity areas: an ethnoarchaeological study of turf buildings at Thverá, northeast Iceland,* 2012. DOI: 10.1016/j.jaa.2011.11.001.

[21] Hnatowich, E. Lamb, and K. Stewart. *Reintroducing vascular and non-vascular plants to disturbed arctic sites: investigating turfs and turf fragments,* 2023. DOI: 10.3368/er.41.1.3.

[22] S. Malleshaiah et al. *Influence of aerification technique on recuperative potential of warm season turfgrasses,* 2018. DOI: 10.56093/ijas.v88i5.80090.

[23] H. Nunome et al. *High load stress-strain property of natural turf for professional use, various types of natural, hybrid and artificial turfs in football,* 2020. DOI: 10.3390/proceedings2020049142.

[24] M. Fontanet, D. Fernàndez-García, and F. Ferrer. *The value of satellite remote sensing soil moisture data and the dispatch algorithm in irrigation fields.* 2018. DOI: 10.5194/hess-2018-94.

[25] J. L. Starr and I. C. Paltineanu. *Methods for Measurement of Soil Water Content: Capacitance Devices,* 2002.

[26] D. D. Bosch et al. *Large scale measurements of soil moisture for validation of remotely sensed data: Georgia soil moisture experiment of 2003,* 2003. DOI: 10.1016/j.jhydrol.2005.08.024.

[27] M. H. Cosh et al. *Temporal stability of surface soil moisture in the Little Washita River watershed and its applications in satellite soil moisture product validation,* 2006. DOI: 10.1016/j.jhydrol.2005.08.020.

[28] H. Koyuncu. *H. Koyuncu.* 2021. DOI: 10.35378/gujs.720778.

[29] H. Quinones and P. Ruelle. *Operative Calibration Methodology of a TDR Sensor for Soil Moisture Monitoring under Irrigated Crops,* 2001.

[30] F. Hu, C. Chen, and X. Li. *Design of wireless water-saving irrigation controller.* 2013. DOI: 10.4028/www.scientific.net/amm.278-280.1577.

[31] A. Abdelfattah et al. *Calibration of soil humidity sensors of automatic irrigation controller.* 2020. DOI: 10.1051/bioconf/20201700249.

[32] M. Abidin et al. *Soil moisture content and density prediction using laboratory resistivity experiment.* 2013. DOI: 10.7763/ijet.2013.v5.652.

[33] H. Eller and A. Denoth. *A capacitive soil moisture sensor,* 1996.

[34] M. S. S. Varma et al. *Design, development and performance study of a polymer coated capacitive sensor for measuring the moisture content of the soil,* 2016.

[35] J. González-Teruel et al. *Design and calibration of a low-cost sdi-12 soil moisture sensor*. 2019. DOI: 10.3390/s19030491.

[36] I. Kulmány et al. *Calibration of an arduino-based low-cost capacitive soil moisture sensor for smart agriculture*. 2022. DOI: 10.2478/johh-2022-0014.

A. Additional Material

This appendix is reserved for final hardware schematics, PCB drawings, firmware listings, raw calibration data, additional field-test plots, and deployment photographs.